



Research Lab Group 1

AI and COVID

Report

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List of Figures.....	v
List of Tables	vi
1. Introduction	1
2. Project Management	4
2.1 Project Leaders	4
2.2 Project Meetings	4
2.3 Project Plan.....	5
3. Research design	9
4. Background Research.....	11
4.1.1 Traditional Expert System Architecture.....	13
4.1.2 Types of Expert Systems.....	13
4.2 Literature of Expert Systems in Covid-19.....	16
4.3 AI and ML Approaches.....	18
4.4 Clustering	21
5. Research Evaluation.....	23
5.1 Questions for Expert System.	23
5.2 Selection of Approach.....	25
5.3 Case Based Reasoning	26
5.3.1 Defining a Case.....	26
5.3.2 Case Structure.....	27
5.3.3 Case Solution.....	31
6. Data Collection and Insights	34
6.1 Data Quality Methodology	34
6.2 Data Collection for Active Cases and Deaths	35

6.3	Data Collection for Weather Data	39
6.4	Data Collection of ICU, Death Cases, Severe Cases, Population, and Average Age.....	42
6.5	Cases Time Frames Collection Methodology	53
6.5.1	Option 1: Observing Changes in The Stringency Index	53
6.5.2	Option 2: Examining Changes in Specific Policy Levels	54
6.5.3	Option 3: Using a Fixed Time Window.....	54
6.5.4	Option 4: Collecting Important Dates.....	55
6.5.5	Cases Time Frames.....	55
6.6	Building Case Library	79
7.	System Architecture.....	82
7.1	Users.....	87
7.2	Business Layer.....	88
7.3	Application Layer	89
7.4	Technology Layer	93
8.	System Implementation.....	94
8.1	Tech Stack	94
8.2	System Flow and User Interface.....	94
8.3	API	98
8.4	Deployment	99
9.	Alternative Approach.....	106
9.1	Rule Based Approach.....	106
9.2	Machine Learning Approaches.....	108
9.3	Rule-Based, CBR and Machine Learning: A Mixed Approach	110
10.	Discussion, Recommendation, and Research Scope	112

11.	Conclusion	114
12.	References.....	115

List of Figures

Figure 1 .3.1 Quarter 3_22 Activities Gnatt Chart.....	6
Figure 1.3.2 Quarter 4_22 Activities Gnatt Chart	7
Figure 1.3.3 Quarter 1_23 Activities Gnatt Chart	7
Figure 1.3.4 Quarter 2_23 Activities Gnatt Chart	8
Figure 3 Design Science Research (DSR).....	10
Figure 4 Typical Expert System Architecture	13
Figure 5: Script for downloading data.....	35
Figure 6 Script for converting JSON to CSV	36
Figure 7 Script for filling missing active cases.	37
Figure 8 Plot of actual active cases and calculated active cases.....	38
Figure 9 System Architecture	83
Figure 10 Recommendation page before recommendation.....	95
Figure 11 Recommendation result	96
Figure 12 Case Builder Page	97
Figure 13 Case Library Page	98
Figure 14 .env file example	100
Figure 15 Gunicorn configuration file	101
Figure 16 service file for the application.....	102
Figure 17 Service status output.....	102
Figure 18 Nginx configuration file	104
Figure 19 Evaluation metrics for KNN and decision tree	110

List of Tables

Table 1 Project leaders	4
Table 2 Project meetings	5
Table 3 Expert systems tabular comparison	16
Table 4 Comparison of various approaches	21
Table 5 Timeframe for cases	79

Abstract

This document presents the architecture of an expert system along with a working prototype specifically designed to assist policy makers and decision makers during a pandemic. The focus of this research is on three key areas: Case-Based Reasoning (CBR), system architecture, and decision-making for policies. The CBR approach is employed to leverage past cases and provide valuable insights for decision making. The system architecture is carefully designed to ensure efficient processing and analysis of data. The prototype showcases the system's flow, user interface design, and the integration of an API. The data collection process ensures high data quality, and visualizations are used to illustrate the relationship between COVID-19 cases and policy measures. This expert system prototype offers a promising solution to support policy makers by utilizing CBR, robust architecture, and informed decision-making processes, ultimately enhancing their ability to respond effectively during pandemics.

1. Introduction

Author: Neil (100%)

The COVID-19 pandemic has had an unprecedented impact on societies and economies worldwide, highlighting the critical need for effective policy making and decision support systems during times of crisis. The global response to the pandemic has been multifaceted, with governments and policymakers grappling to balance public health concerns, socioeconomic implications, and the urgent need to mitigate the spread of the virus. As a result, there is an increasing demand for advanced technological solutions that can assist policy makers in navigating the complexities of future public health emergencies.

Artificial Intelligence (AI) has emerged as a powerful tool with the potential to revolutionize decision-making processes in various domains. Its ability to analyze vast amounts of data, identify patterns, and derive valuable insights has opened up new possibilities for addressing complex challenges. In the context of the COVID-19 pandemic, AI-based approaches have been instrumental in understanding the spread of the virus, predicting its impact on healthcare systems, and assisting in the development of effective mitigation strategies.

This research internship aims to contribute to the field of AI and public health by developing an expert system architecture that harnesses the power of AI to understand past COVID-19 data and provide informed recommendations for policy makers. By leveraging the wealth of available data from the pandemic's progression, this system will aid in decision-making processes, offering valuable insights into the best course of action to be taken in the event of a similar crisis in the future.

The primary objective of this research is to design the architecture of an expert system that can effectively analyze historical COVID-19 data, including infection rates, mortality rates, vaccination rates, deaths, weather, and various other relevant factors. By integrating Case Based Reasoning with the help of similarity techniques like Euclidian distance, A working prototype that can identify patterns within the data was created from the designed architecture, thereby enabling policy makers to make data-informed decisions in real-time.

To achieve this objective, the project will follow a systematic research approach that includes data collection, preprocessing, feature engineering, and model development. Various machine learning

techniques and approaches will be explored to develop a robust and interpretable system. The resulting expert system will serve as a valuable decision support tool for policymakers, providing them with actionable insights to shape effective and evidence-based strategies during future public health crises.

In conclusion, this thesis presents an ambitious endeavor to develop an AI-powered expert system that harnesses past COVID-19 data to inform policy making during similar crisis situations. By bridging the gap between AI and public health, this research strives to contribute to the growing field of AI in healthcare and provide tangible benefits to society. Through the creation of an intelligent decision support system, policy makers will be equipped with the necessary tools to navigate future challenges, mitigate risks, and ensure the well-being of communities worldwide.

This research also aims at answering the following research questions:

1. How can decision makers better allocate resources in the event of another outbreak of the Covid-19 virus?
2. What would be an optimal policy that could be implemented in the event of another outbreak of the Covid-19 virus with considerations to Lockdowns as well as Mask & Vaccination policies.

This project handbook outlines the research and development process of an expert system powered by Artificial Intelligence (AI) and Machine Learning (ML) techniques. The handbook is divided into several sections, each addressing a specific aspect of the project.

The Background Research section provides an overview of the research design and conducts an extensive literature review. It explores the concept of expert systems, including their architecture, types, and a comparative analysis. Additionally, different AI and ML approaches are discussed, showcasing their potential applications in the context of policy making. The section concludes with a paper discussion and an outline summarizing the findings.

The Research Evaluation section focuses on evaluating the effectiveness of the expert system. It poses relevant questions that the system should address and selects specific approaches for further investigation.

The Data Collection and Insights section explains the methodology used to ensure high-quality data. It identifies the necessary data to answer strategy-related questions and provides a

comprehensive list of data sources. The section also showcases the code used for data collection and presents visualizations that illustrate the relationship between COVID-19 cases and implemented policies.

The Case-Based Reasoning section offers an overview of the underlying approach, defines the structure of cases, and explains how the system learns from past cases to make informed decisions.

The System Architecture section provides a detailed explanation of the expert system's architecture, including its components and their interactions, highlighting how the system effectively analyzes and processes data.

The System Implementation section describes the implementation of the expert system, outlining the system flow, user interface design, and the potential integration of an API to enhance its functionality.

The Discussion, Recommendation, and Research Scope section critically evaluates the research outcomes, identifies any limitations, and offers recommendations for future work. It emphasizes the significance of the achieved results and highlights potential areas for further exploration and improvement.

2. Project Management

2.1 Project Leaders

Team Leaders	Duration	
	From	To
Neil & Rahul	28.10.2022	21.11.2022
Shehab & Sudan	22.11.2022	18.01.2023
Nishan & Shuvam	19.01.2023	07.03.2023
Asnel, Prantik & Prajwal	08.03.2023	Present

Table 1 Project leaders

2.2 Project Meetings

Official Meetings

No.	Purpose of Meeting	Date
1	Presentation of Initial Research Pitch	06-10-2022
2	Presenting Project Objectives and Related work	28-10-2022
3	Case Based Reasoning, Planned Architecture	21-11-2022
4	Required Data, Case Based Reasoning, Planned Architecture	21-12-2022

5	Data Pipeline, Phases, Data Sources and Reason for Choosing Data	18-01-2023
6	Data Conceptualization and Dummy Cases	06-02-2023
7	Defining Cases, Shift to Policy Making as Research Objective	07-03-2023
8	Definition & Structure of Cases, Updated Project Plan	22-03-2023
9	Prototype implementation	05-05-2023
10	Final Presentation	09-06-2023

Table 2 Project meetings

2.3 Project Plan

Author: Prantik (100%)

Project Plan plays an essential role to provide us to allocate tasks and division of resources. Initially, there were three teams: The Data Source team handled the collection of data sources, creating datasets and time frames, along with text mining and topic modeling approaches. Further the Architect team members were able to identify the process flow of the entire system resulting in desired solutions.

The next task of the project leaders was to come up with a project plan, where the project leaders had to define deliverable and milestones with the initial scope of project. The project teams conducted several informal meetings throughout the duration of the project for better collaboration and streamlined approach in project planning and implementation. The project leaders arranged formal meetings with the project supervisors to present the ongoing iterations and gather feedback so that final deliverable of the research project could be improved.

All members of the project first had to understand the needs of the project by. The data source team performed data collection and the Development team researched machine learning methods in python. After getting an understanding of CBR methodology, the team was able to implement this gathered knowledge for the data processing task on the data collected. The creation of a simple GUI prototype is one of the milestones of the project which was achieved remarkably in a short duration of time.

Most of the project time was spent on exploring and implementation of the different approaches.

Thereafter, based on the requirements and understanding, the data source team collected more data for the training purpose of the model.

One of the major deliverables of the project plan was the final presentation wherein the team presented the architecture and the implementation of prototype that we achieved in our research project. The whole research process throughout the research lab has been documented in the research handbook.

Gantt Chart:

The Gantt Chart has been generated using Microsoft Project Plan.

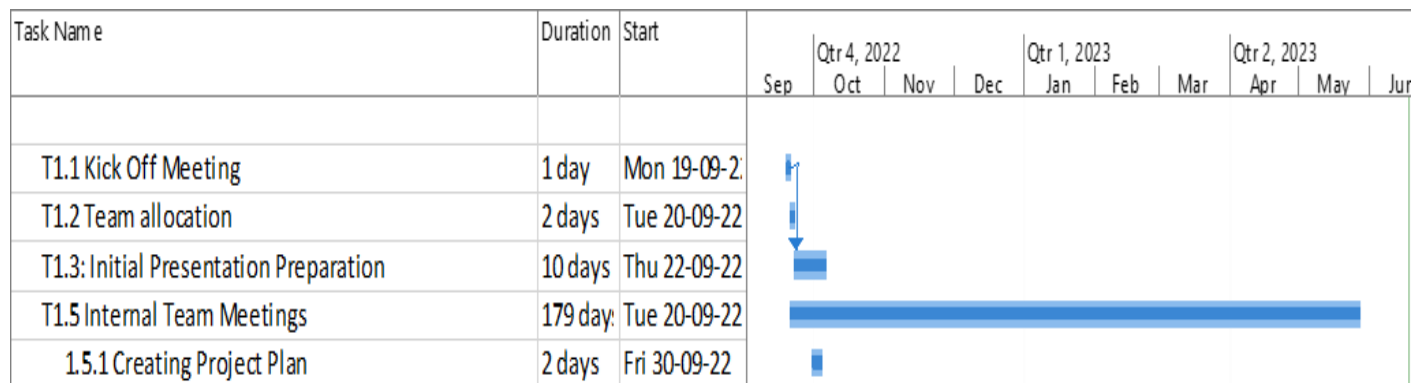


Figure 1 .3.1 Quarter 3_22 Activities Gnatt Chart

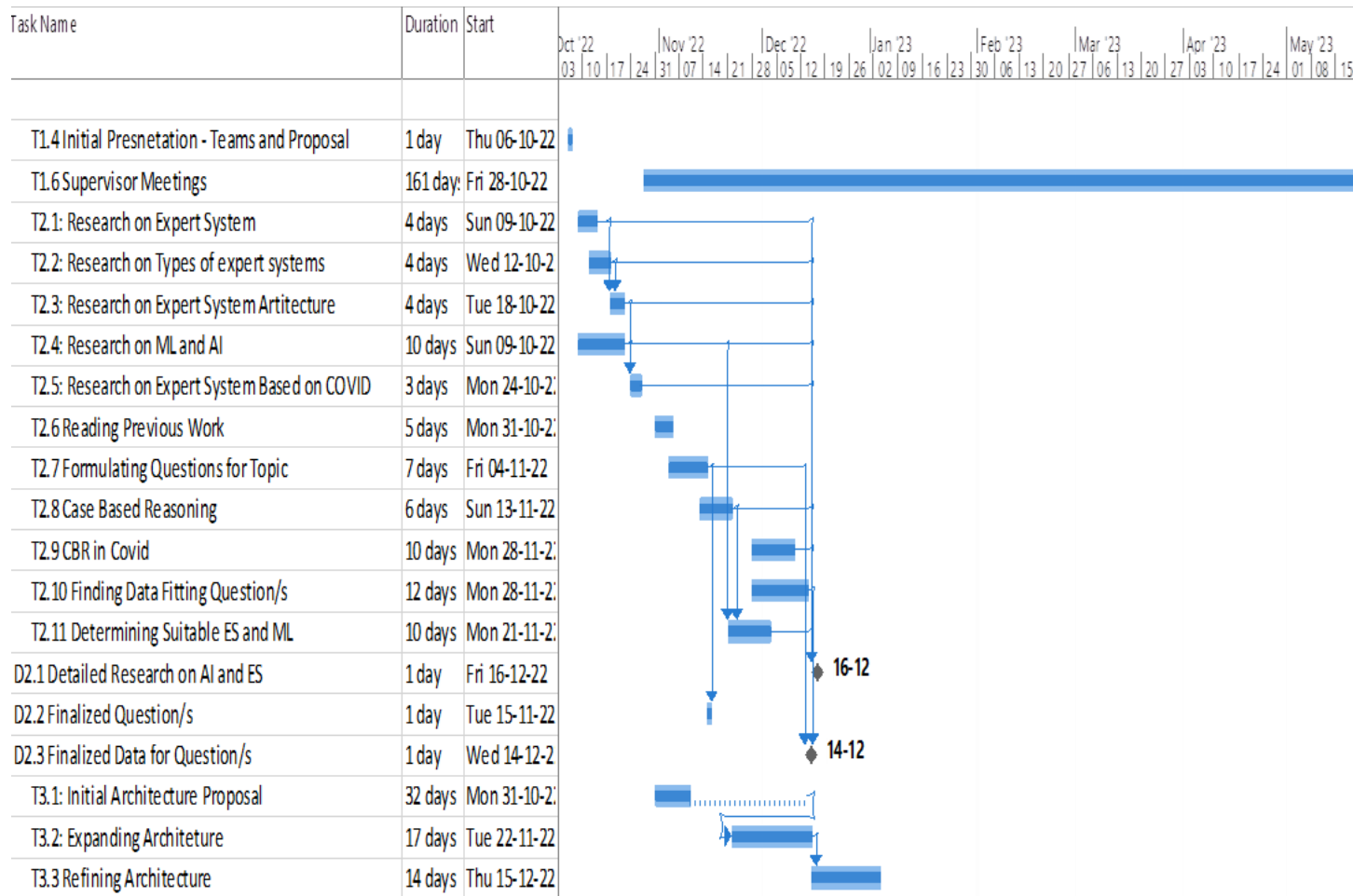


Figure 1.3.2 Quarter 4_22 Activities Gantt Chart

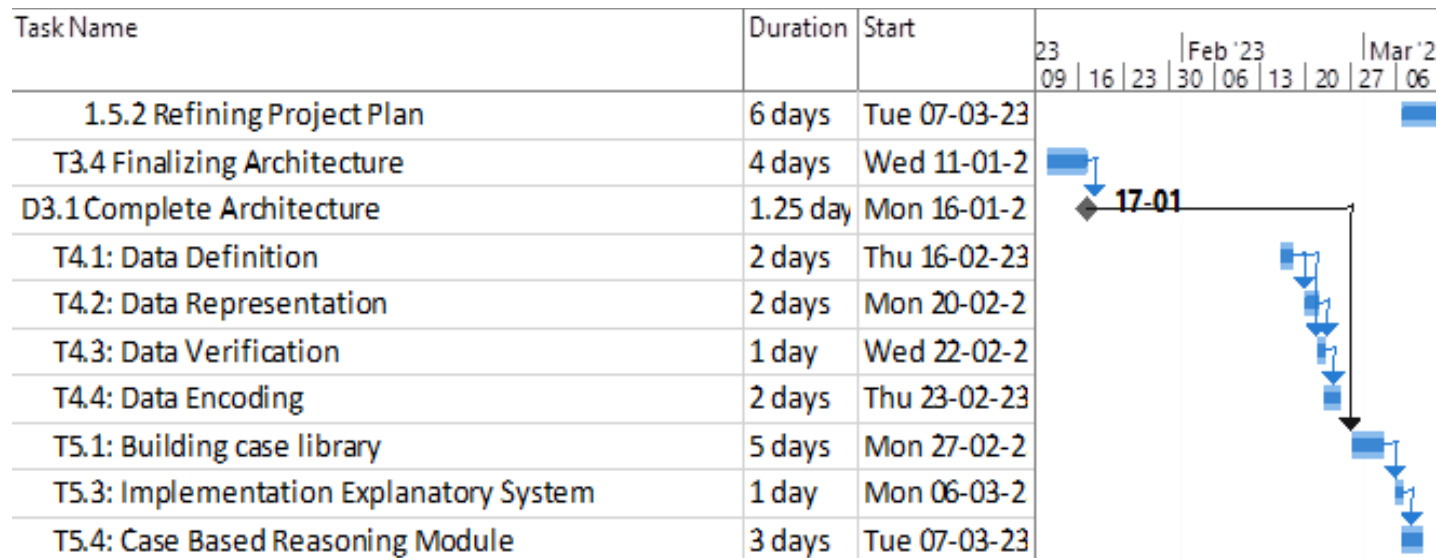


Figure 1.3.3 Quarter 1_23 Activities Gantt Chart

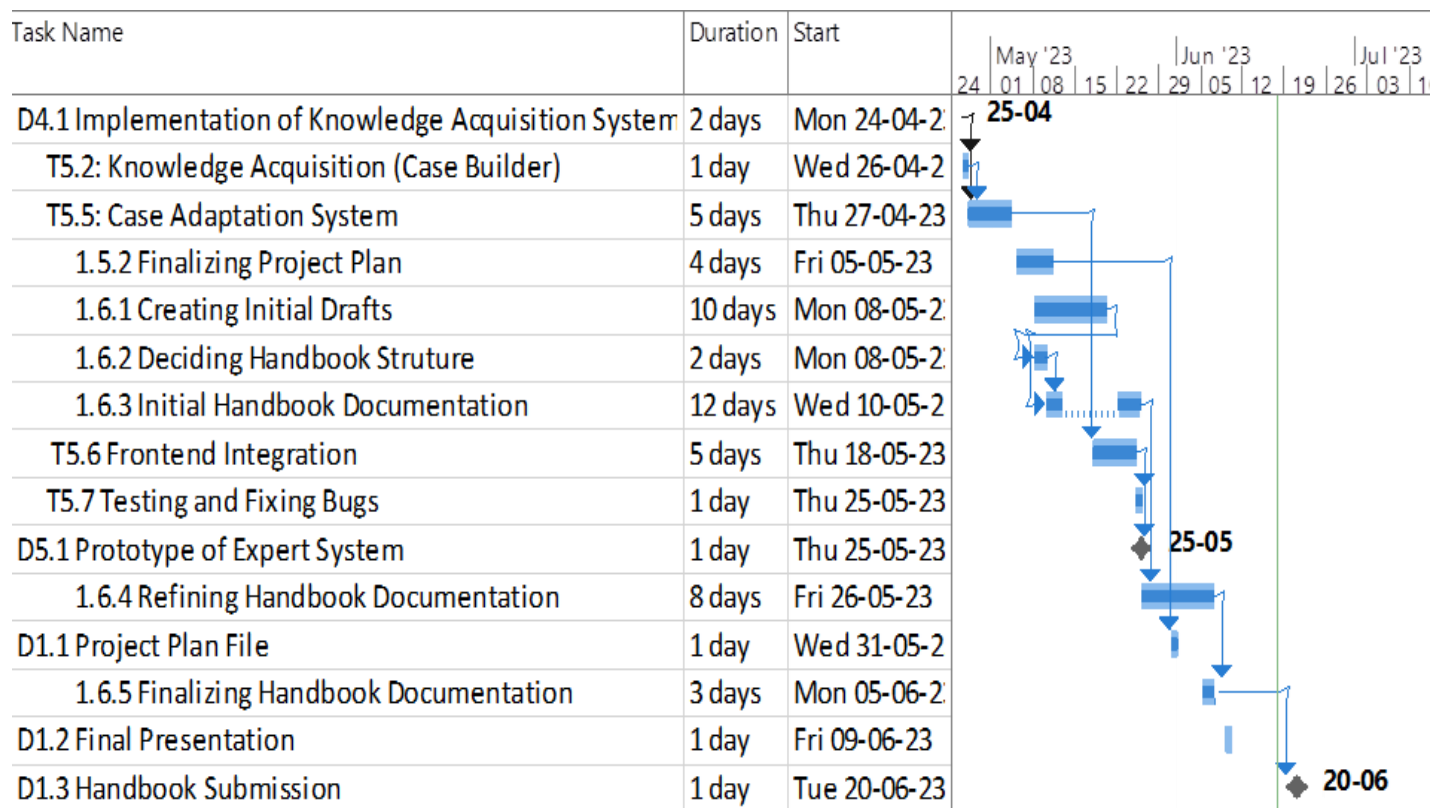


Figure 1.3.4 Quarter 2_23 Activities Gantt Chart

3. Research design

Author: Christian Asnel (100%)

We opt for the Design Science Research (DSR) from Vaishnavi & Kuechler as our guiding research method [1]. This decision stems from our recognition of DSR's profound relevance and applicability in terms of addressing the unique demands and objectives of our studies.

The essence of DSR is its commitment to the creation and critical evaluation of practical artifacts, which can encompass various entities such as systems, models, and theoretical frameworks. These artifacts, forged in the crucible of research, serve as instrumental tools to tackle real-world problems. They not only contribute to the enrichment of our understanding but also facilitate the development of robust solutions that address these problems effectively.

A noteworthy characteristic of DSR that resonates with our research approach is its inherent flexibility. This trait is manifest in DSR's embrace of an iterative process. The cycle of development and evaluation that DSR espouses allows for continual refinement of the artifacts. Such a dynamic approach enables us to improve and innovate persistently, thereby allowing our research outputs to remain adaptable and responsive to the evolving realities of our study domains.

Furthermore, DSR encourages a collaborative ethos, fostering a cross-pollination of ideas and strategies. It invites researchers to engage in shared dialogue and intellectual exchange, thereby enhancing the quality of the research output.

Considering the mission of our research lab, which is rooted in addressing practical issues and contributing valuable solutions, we find DSR's methodology to align seamlessly with our objectives. Its pragmatic approach, iterative structure, and collaborative spirit align well with our aim to generate research outputs that are both theoretically rigorous and practically relevant.

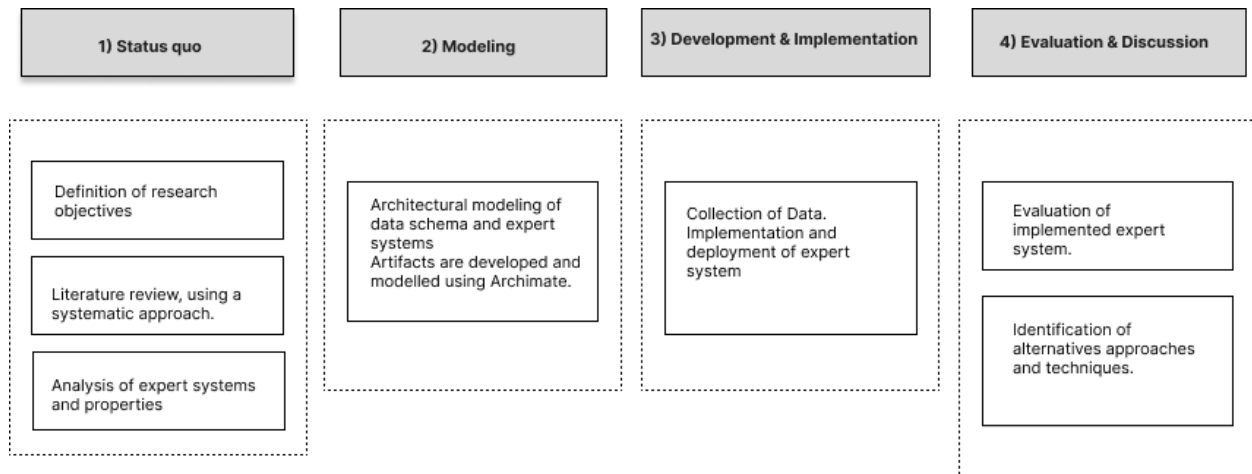


Figure 5 Design Science Research (DSR)

During our research lab, we iteratively move through the depicted phase to build on to our result. We use Archimate to model our architectural artefact.

4. Background Research

In this section we explore expert systems, including its architecture and the different types, literature on expert systems in relation to Covid-19 and discuss various machine learning Expert Systems approaches and algorithms.

Author: *Shehab (30%), Shuvam (70%)*

Humans are capable of acquiring and accumulating knowledge whether that is generally in life or in a specific field [2]. Some of us specialize in fields and as the years pass, we become experts in our field of choice. Expert systems attempt to imitate a human's knowledge to make decisions where a user can interact with it to address a problem [3]. In other words, acting like a human expert to make a decision. An expert system is defined in the literature as the ability of a computer system to solve problems based on skill or skills also known as knowledge of one or more experts [4]. The field of expert systems is based on and part of Artificial Intelligence [5]. Intelligence can be defined as one's ability to acquire knowledge to solve a problem. Artificial Intelligence on the other hand, is the ability to produce cognitive intelligence artificially, usually through machines to solve problems [6].

On his book "The Science of the Artificial", Herbert A. Simon, a Nobel Prize-winning economist and computer scientist who was one of the pioneers of artificial intelligence, opine that, "The expert system is not a panacea, but it can be a very useful tool. It can help us to extend our intellectual abilities and provide us with new insights into complex problems." Simon's believed that expert system could aid us in expanding our intellectual horizons and giving us fresh perspectives on challenging issues by capturing the knowledge of specialists in a particular sector. In other words, expert systems are not intended to replace human experts, but rather to supplement and make more accessible their knowledge.

At Stanford University in the early 1960s, Joshua Lederberg and Edward Feigenbaum created the first expert system, known as Dendral [7]. The system's purpose was to recognize the structures of organic molecules and decipher complex chemical spectra. Previously, completing this task needed a tremendous deal of knowledge and experience and took a long time. The fact that Dendral could complete this task faster and more precisely than human experts represented a significant advance for artificial intelligence.

The inability to accurately capture and portray human expertise in a form that a computer could comprehend was one of the early difficulties in constructing expert systems. Due to this, methods for "teaching" expert knowledge to computers in an organized way—known as "knowledge engineering techniques"—were created.

Following their original creation in the 1960s, expert systems saw an increase in their application in industry and government, particularly in areas such as finance, manufacturing, and defense. During the 1970s and 1980s. Expert systems became increasingly potent and capable of handling larger and more complicated datasets as computer hardware and software advanced.

MYCIN was one of the first effective expert systems, created at Stanford in the early 1970s to aid in the diagnosis of bacterial illnesses. MYCIN set the groundwork for the creation of numerous more expert systems in a variety of domains by achieving diagnosis accuracy equivalent to that of human experts [8]. In the later years, expert systems became more widely used in wide verity of sectors.

Expert systems, however, have come under scrutiny for their shortcomings, including their inability to learn from experience or adjust to changing scenarios. As a result, machine learning and deep learning became the main areas of concentration for AI development. Knowledge representation also became a key ingredient for the development of more powerful, flexible, and domain-specific expert systems, and has enabled them to tackle a wide range of complex problems in different domains. Over time a variety of knowledge representation techniques have been used in expert systems, including rule-based systems, semantic networks, and logic-based systems. The choice of technique depends on the nature of the problem and the available data and expertise [9] [10].

Natural language processing, machine learning, and deep neural networks are used by contemporary expert systems like IBM Watson, Amazon Alexa, Google Assistant, and DeepMind AlphaGo to provide insights, make suggestions, and carry out difficult tasks. Amazon Alexa [11] and Google Assistant [12] are voice-activated assistants that can complete a variety of tasks, while IBM Watson [13] is a question-answering system that has been employed in a variety of businesses. DeepMind AlphaGo is an expert system that, using deep neural networks and machine learning, has mastered the challenging board game of Go [14]. We will probably see much more advanced and potent expert systems in the future as AI and machine learning technology develop.

4.1.1 Traditional Expert System Architecture

Authors: Neil (50%), Shuvam (50%)

Expert systems are knowledge systems that use computers as a tool and draw upon expertise and knowledge outcomes to recognize and address the problem at hand [15]. Expert systems are primarily made up of four integral parts: the knowledge base, the inference engine, the knowledge acquisition component, and the explanation interface.

- **The knowledge base** represents all the facts and rules of the system. It consists of domain knowledge, procedures that are relevant to the domain as well as rules necessary to solve complex problems.
- **The inference engine** acts as the interpreter of the system. It is responsible for gaining facts from the knowledge base, proposing solutions to the problem at hand as well as offering explanatory and debugging capabilities.
- **The knowledge acquisition** component is responsible for gaining more knowledge from various sources and adding it to the knowledge base.
- **The explanation interface** oversees explaining the solution that it proposes to the user while also explaining how it came to such a conclusion.

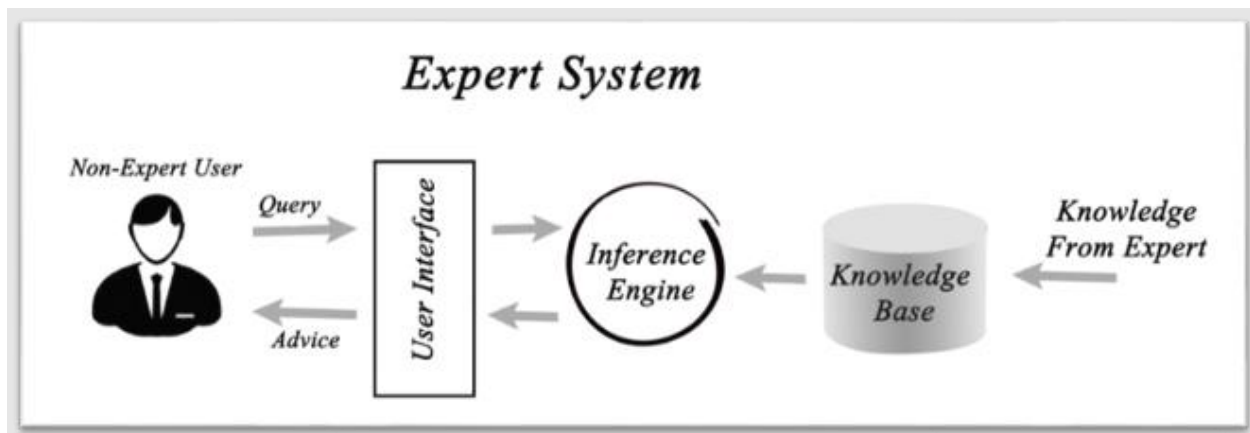


Figure 6 Typical Expert System Architecture

4.1.2 Types of Expert Systems

Authors: Shehab (50%), Sudan (50%)

As discussed, an expert system in general can make decisions depending on the field specified and the knowledge given. These decisions are based on the information provided to the system. The

information is supplied by the experts and is transferred and stored in what is referred to as a knowledge base which includes facts and rules [4]. According to Maad and Rana [16], the decision-making process is derived from an inference engine which can deduce a result based on the facts and the rules stored in the knowledge base and from user input. These decisions are made in an “IF THEN” fashion also known as rule-based decisions, where the rules are described in the left-hand side and the actions are in the right-hand side. For example, let A of n represent conditions and B represent an action to take, IF: A_1 , and A_2 THEN: B . In this simple case, if a user were to input A_1 and A_2 , then the expert system will respond with B because B is the result of the conditions A_1 and A_2 . However, there are different types of expert systems that use different inference strategies other than rule-based which include Fuzzy-based, Neural, Neuro-Fuzzy, frame-based expert systems and case-based reasoning [17] [18]. Below we present a list of these types of expert systems [19].

- **Rule-based expert system:** according to [18] in a rule-based expert system, knowledge is represented in the form of rules, expressed as IF-THEN in the knowledge base. The inference engine derives results by processing the rules. However, it does not have the ability to learn because it requires constant updates to the knowledge base. A rule-based expert system is difficult to maintain but has an explanation facility due to the rules. It also takes time to process because each one of the rules needs to be evaluated and processed.
- **Fuzzy-based expert system:** according to [19] in a fuzzy-based expert system the knowledge to represent is usually vague and unclear. For example, an expert may express an opinion like the following “likely heavy,” “very small” and “slightly overwhelmed.” The point is fuzzy based system usually focus on range or scale to reach a conclusion unlike rule-based which is mostly binary. This can be represented using fuzzy logic, which is a type of logic that explains fuzziness in data. A fuzzy-based expert system cannot learn, but it can explain its decisions based on linguistics. The inference engine processes rules using a technique called fuzzification and defuzzification in a four-step process, fuzzification, rule evaluation, aggregation of rule outputs and finally defuzzification. However, its processing time is less compared to rule-based ES. It is worth noting that fuzzy-based expert systems also use IF-THEN to reach conclusions.
- **Frame-based expert system:** according to [19] in a frame-based expert system knowledge is represented in frames. A frame is simply a data structure which includes specific knowledge.

A frame consists of attributes and names. In this way knowledge can be captured in an object. The inference engine in this system mainly tries to find a matching attribute to draw a decision. This system doesn't have the ability to learn but it has an explanation facility, and it is easier to maintain compared to the previous systems. Its processing time is also very fast due to its frames structure.

- **Neural-based expert system:** A neural-based expert system is a type of artificial intelligence that uses neural networks, which are a type of machine learning model inspired by the structure and function of the human brain [20]. According to [19] in a neural-based expert system knowledge is represented in the form of IF-THEN but in neurons and weights much like a neural network in AI. However, unlike previous systems, it can learn due to its nature. It also has an explanation facility, and its inference engine uses a three-layer neural network which is combined with rule extraction. It is also easy to maintain. The processing time in this type of system is fast but the learning can take time.
- **Neuro-fuzzy expert system:** according to [19] it is a combination of fuzzy and neural expert system where the advantage here is to take the learning and parallel computation abilities from neural expert systems and the knowledge representation of fuzzy based expert systems. This means knowledge is represented in linguistic variables and IF-THEN but in a fuzzy way. Neural networks are machine learning models inspired by the structure and function of the human brain, while fuzzy logic is a type of reasoning that deals with uncertain or vague information. Hence, it can learn due to the neural network combination, as well as having an explanation facility. Its inference engine follows the fuzzification and defuzzification process. However, it is difficult to maintain due to the fuzzy nature, but the processing time is fast.
- **Case-based expert system:** according to [21] these systems represent knowledge in terms of cases. This means that the system will have historical cases and a target case. The target case is what the system is trying to find a solution for. These systems can be learned depending on the approach taken. The way the inference engine or the reasoner works is that a target case is given as an input, the reasoner finds matching historical cases and gives the most relevant solution to the case at hand.

Below is a tabular comparison of the different approaches to expert systems as has been discussed in earlier sections.

Attributes	Rule-based	Fuzzy-based ES	Frame-based	Neural-based	Neuro-fuzzy-based	Case-based
Knowledge representation	IF-THEN rules	Based on scales using fuzzy logic	Structured frames	IF-THEN rules in the neural knowledge base	Linguistic variables and IF-THEN in fuzzy structure	Cases
Learning ability	No	No	No	Yes	Yes	Yes
Explanation	Yes	Yes	Yes	Yes	Yes	Yes
Inference engine	Process rules and reaches conclusions.	Process rules using fuzzification and defuzzification	Search for matching attributes	Neural network and rule extraction	Fuzzification and defuzzification	Comparing historical cases with target case

Table 3 Expert systems tabular comparison

4.2 Literature of Expert Systems in Covid-19

Authors: Shehab (50%), Sudan (50%)

In this section we present various systems and approaches proposed for Covid-19 including the usage of expert systems and machine learning in some proposed work.

Expert systems have proven their value over the years; hence they have been used in many areas including monitoring, diagnosis, scheduling, planning, learning, design, and explaining [4] [22].

In fact, amongst the very first Expert Systems was DENDRAL which was used to determine the

structure of chemical molecules and investigate how scientists come up with and discover novel hypothesis [23]. As this research relates to AI and COVID specifically with Expert Systems, there has been research and projects that have implemented Expert Systems to solve COVID related problems.

Salman et al [24], in their paper Expert System for COVID-19 Diagnosis, have presented an Expert System which helps doctors to detect or diagnose COVID-19 symptoms, given a body of knowledge such as fever, cough, pneumonia and other factors. COVIDC is another example of the Expert System used in the context of COVID-19 to diagnose and predict the severity of COVID based on a chest CT scan. It uses a two-phase classification approach in Machine learning with deep features map with supervised classification algorithms like SVM and random forest [25]. Another Expert System was proposed by Goita and Sidibe to provide analysis, treatment and vaccination options for COVID-19 which was aimed to help health workers and decision-makers to combat COVID-19 [26]. In the paper “Putting the world back to work” they mentioned the expert system General Guide (GG) their objectives were to build an expert system that helps slow down the spread rate of COVID-19, forecast and ensure the safety of people. For this, it uses the ML algorithms like support vector machine, Decision tree, and Neural Networks [27]. Duan and Jiao [28] in their paper “Novel Case-Based Reasoning System for Public Health Emergencies” have proposed a system that helps in infectious outbreaks. The main problem they were trying to solve is that with infectious outbreaks certain knowledge does not exist yet and it could take time to gather and collect that knowledge to make decisions. With this insufficient knowledge decision-makers can find themselves in a tricky situation. The authors suggested that learning from historical cases can be helpful as it most likely results in a similar course of actions. Their study compares different reasoning systems such as case-based, rule-based, and model-based reasoning systems. They concluded that case-based reasoning excels at solving problems where there is a lack in domain knowledge and can allow for incremental knowledge. The proposed system is based on CBR which uses three steps or phases to reach a conclusion. The first phase is to calculate the similarity between current and historical cases. Secondary, grey clustering analysis is applied to retrieve the most relevant cases. In the last phase, the proposed system uses cuckoo search algorithm to find the best solution for the given case. The study gave an example of the system where it applied a problem of an outbreak to give a solution of best course of action. The example included factors in the problem such as infection ratio, number of infections, average quarantine

period etc., and the system suggests the best actions to take such as the number of health workers, beds, experts needed etc.

Aimed to prevent and control COVID-19 in urban rail transit with **emergency responses**, Yuning et al. have **proposed** an expert system based on Cased based reasoning. In this method, historical cases were used, and target cases were used to find out the similarities between them. And then the comprehensive utility value of the emergency response of each similar case is calculated by taking the reference of the utility evaluation of the emergency response effect and response cost of each similar historical case. The data needed for these studies were infectivity, routes of infection, pathogenicity, propagation distance, and urban population density. For the evaluation part, the decision makers take suggestions from the experts in the field of emergency response where they took the five evaluation indexes and calculated the weighted vector for each of them. And the suggestion from the experts was combined to get the emergency response effect and response cost. The examples provided for the emergency response were sanitization of vehicles, strict entrance in airports, and stations, temperature check while entering a premise, etc. Therefore, the decision-makers will choose the right action based on the previous similar case situation which was effective in preventing and controlling COVID-19 [29].

In the paper “Fuzzy case-based reasoning approach for finding COVID-19 patients priority in hospitals at source shortage period” Selvaraj **et al.** have mentioned the fuzzy cased-based reasoning approach for making decisions on prioritizing patients based on their severity. The mortality of patients was increasing and one of the main causes was an improper allocation of resources to the needed one. Cased Based reasoning has shown a vital part in making decisions, especially in the medical domain with a lot of uncertainty in knowledge. CBR is a valuable method for diagnosis of the disease with knowledge-based techniques and to solve the problem by searching the similar cases from the past and reusing it. The data taken into consideration were symptoms, age, duration of symptoms, oxygen saturation, blood pressure, chest x-ray, and comorbidities (other medical conditions). As a result, the system will show the calculated data considering all the factors mentioned above and health professionals can make the decision based on the given output. [30].

4.3 AI and ML Approaches

Authors: *Rahul(50%), Prajwal(50%)*

The COVID-19 pandemic has brought about unprecedented challenges for healthcare systems worldwide. In response, researchers and data scientists have been exploring various machine learning algorithms and techniques to improve the accuracy of COVID-19 detection and forecasting. This passage discusses some of these techniques, including Convolution Neural Networks (CNN), Support Vector Machines (SVM), Random Forest, Regression, XGBoost, and Autoregressive Integrated Moving Average (ARIMA). These techniques have been used to analyze COVID-19 data and provide insights into the spread and impact of the disease. CNN has been used to improve the accuracy of COVID-19 X-ray image classification, while SVM has been used to detect COVID-19 from symptoms. Random Forest has been used to forecast the impact of COVID-19 and track the geographical location of individuals. Regression has been applied to assess the impact of active cases due to daily positive cases. XGBoost has been used to boost decision trees and avoid overfitting, while ARIMA has been used to forecast events over a period of time. These techniques have the potential to aid in the fight against COVID-19 and improve our understanding of the disease.

- **Convolution Neural Networks:** CNN with extra layers to improve the COVID-19 X-ray image classification accuracy. In neural networks, the CNN structure is specially designed to process the two-dimensional image tasks although it can also be used in one- and three-dimensional data. In binary classification we have only one output node needed to classify the data to one of the two given classes.

To train the CNN model, first, the dataset has been obtained, the dataset obtained for training the model was very small in size and imbalanced, it has been extended using data augmentation techniques to increase its size and to make the model training feature rich image flipping and rotation at different angles have been used to generate more data. For dataset balancing in terms of proportion of images with different class labels, the dataset has been further extended with some more image instances of the minority class. After data augmentation and dataset balancing, the CNN model has been trained using a total of 800 images (400 COVID-19 and 400 normal) and then the model has been tested by using a test set. [31]

- **SVM:** SVM is used in the detection of COVID-19 from the symptoms because of the uncertainty of the data. A multi criteria data can be created with the symptoms of COVID 19, based on the infected symptoms infected status can be found out. Three classes, which are not

infected, mildly infected, and severely infected. SVM is chosen because it uses kernel trick to convert low-dimensional input space to high-dimensional space and thus converts the Non separable problem to separable problem. [32]

- **Random Forest:** This method has been used to create a time series to forecast the impact of Covid-19 for daily or weekly intervals. This model has also been used to track the geographical location of the individual along with health and travel details to predict the possibility of contracting Covid-19, recovery, and death. [33] Multiple decision trees were implemented and the average cumulative output of all the trees is used to create the final projection. Additionally, higher accuracy can be attained by using booster algorithms such as AdaBoost which consider the weights of the individual decision trees.
- **Regression:** A correlation analysis is performed between the various attributes involved to understand how they relate to each other. Later, regression is applied to assess the impact of active cases due to daily positive cases. [34] In most cases, regression relies on more than just one attribute/variable; hence, Multiple Linear Regression is the most preferred approach. The slope and intercept are determined most commonly by the least square method, to fit the regression line and the R-Squared score is used to determine how well the data fits the regression line.
- **XGBoost:** XGBoost is a machine-learning algorithm that helps boost decision trees. Each tree boosts attributes that led to misclassification of the previous tree. XGBoost helps avoid overfitting and deals with missing values effectively. It also works accurately on small datasets but also scales well for large datasets and can be run in parallel on several computers
- **AutoRegressive Integrated Moving Average:** ARIMA model is best used to forecast events over a period of time. This model has already proven to be effective when used to predict the rate of spread of diseases such as dengue and malaria. This model is used for predicting future values based on its own past values while also accounting for changing trends, periodic variation, and disturbances. [35] [36] The use of auto-regression and moving average eliminates trend and/or seasonality. The average errors in both the prediction of the number of cases and deaths were lower for the ARIMA model than the XGBoost Model. [36]

Pros and Cons of The Different Methods

Author: Rahul (100%)

Method	Advantage	Limitation
Random Forest	• Efficient on a huge dataset Flexibly incorporate missing information from the past hub of the tree	• High computational cost • Hard to interpret. • overfitting
Decision Trees	• Does not need standardization of information. • Very instinctive and simple to clarify in specialized terms. • Does not need scaling of information also.	• Inadequate for applying regression and predicting continuous values. • Relatively expensive as the complexity and time took are more.
SVM	• More successful in high dimensional spaces. • Relatively memory effective.	• Not suitable for large data sets. • Does not perform very well when the data set has more noise.
Linear Regression	• Simple to actualize and simpler to decipher the yield coefficients. • Independent and ward factors have a straight relationship, this calculation is the best. • Linear Regression is helpless to over-fitting by utilize some dimensionalities decrease methods, regularization (L1 and L2) procedures, and cross-approval.	• Outliers can have huge effects on regression and boundaries are linear in this technique. • Diversely, linear regression assumes a linear relationship between the dependent and independent variables. • Linear regression is not a complete description of relationships among variables.

Table 4 Comparison of various approaches

4.4 Clustering

Author: Christian Asnel (100%)

Clustering is a method in data analysis where data points are grouped together based on certain distinctive features or similarities [37].The process of clustering can yield data divided into

completely separate sets, a concept known as hard clustering, or in overlapping groups, a phenomenon termed as soft clustering [37].

Clustering algorithms are typically classified based on their methodological approach, as follows:

- **Hierarchical clustering:** This technique involves either the aggregation of smaller clusters into larger ones or the division of larger clusters into smaller ones, thereby creating a dendrogram that depicts the relationships among clusters [38].
- **Partitional clustering:** This method divides the dataset into disjoint clusters, and it involves the iterative optimization of a particular criterion function that could represent the local or global structure of the data [38].
- **Density-based clustering:** This type of clustering groups neighboring data points based on certain density metrics, with the DBSCAN algorithm being a prime example of this approach [39].
- **Grid-based clustering:** This method operates under the assumption that the space is partitioned into a finite number of cells, with all operations being performed within this divided space [40]

In this chapter we discussed expert systems, their types and provided literature of expert systems used in the context of Covid-19. We also discussed different AI and machine learning algorithms along with their pros and cons as well as clustering. In the next chapter, we make a list of questions for the expert system to answer, choose one question for the expert system, select an approach to build our expert system, and decided on the data required to build such system.

5. Research Evaluation

In this chapter we discuss and propose various questions that the expert system can answer, select an approach to build the system and its architecture based on previous chapter's research as well as discuss the selected approach in depth.

5.1 Questions for Expert System.

Author: Prajwal(45%) , Rahul(20%) , Prantik(35%)

In managing the COVID-19 pandemic and its associated challenges, selecting the questions for the expert system is crucial. It is essential to implement efficient and effective ways to maximize resource allocation, ensure proper medical care, and reduce the impact of the virus on individuals and communities as the world struggles to deal with the current tragedy.

It is crucial to determine whether booster doses are required for specific populations after considering the necessity that aids in identifying hotspots and establishing various monitoring levels, along with developing variations and probable declining immunity over time. Decision-makers can decide whether extra doses are necessary based on infection rates among those who have received the vaccine, improving overall protection against the virus. As a result, authorities can improve understanding of the severity of the outbreak and recession, they can take measures to lessen the hardships experienced by the pandemic.

These questions are designed to address key areas of concern in managing COVID-19. By considering optimal resource allocation, vaccination strategies, continuous versus risk-based measures, risk assessment for travelers, and providing relief aid, we can successfully navigate the challenges posed by the virus. By using an expert system, we can contribute to evidence-based decision-making and mitigate the impact of the pandemic on individuals, communities, and nations.

- **How can resources, such as medical equipment and medical workers, be optimally allocated to severely affected regions to avoid equipment shortage?.**

Medical equipment or supplies should be distributed in a way that prioritizes regions that are experiencing a severe impact from Covid-19, in order to prevent a shortage of resources in those

areas. The goal is to ensure that the areas that need the resources the most have access to them and that resources are not wasted in areas that do not need them urgently.

- **How can we identify the hotspots and provide various monitoring levels?**

Hotspots are likely to be areas where a particular event, such as an outbreak of a disease, is occurring at a higher rate or with greater intensity than other areas. By identifying these hotspots and providing various levels of monitoring, it allows us to quickly detect any changes or new issues that may arise in the area and respond to them in a timely manner.

- **Should there be a booster dose drive based on the infection rate of vaccinated people?**

Whether or not a coordinated effort should be made to administer additional doses of a vaccine to people who have already been vaccinated, based on the rate of infection among those who have received the vaccine. A booster dose drive may be necessary if there is a high rate of infection among people who have been vaccinated, indicating that the vaccine is not providing adequate protection. The decision to implement a booster dose drive would be based on a number of factors, including the rate of infection among vaccinated people, the prevalence of variants of the disease and the availability of booster doses.

- **Describe which covid-19 strategies should be always used versus only at times of increased risk?**

Determining which strategies to prevent and control the spread of COVID-19 should be implemented consistently and continuously, and which strategies should be implemented only during periods of increased risk. Strategies that should be used at all times include basic infection prevention and control measures. These strategies are effective in preventing the spread of the virus in all settings and should be consistently implemented to protect people from COVID-19. Strategies that should be used only at times of increased risk include measures such as increased lockdown, contact tracing, and isolation of individuals who have been in contact with a confirmed case.

- **Can a risk profile be created for travelers to allow permits to the country?**

Assessing the risk of travelers entering a country and determining whether or not they should be granted entry. This is typically done in order to prevent the spread of a disease, such as COVID-19, into a country. To create a risk profile for travelers, officials would likely gather and analyze

information about the traveler, such as their recent travel history, COVID-19 test results, vaccination status, and any other relevant information. This information would be used to determine the travelers' risk level of contracting or spreading COVID-19. Based on this risk profile, officials would then decide whether to permit the traveler to enter the country. Travelers who are deemed to pose a low risk of contracting or spreading the disease may be granted entry, while those who are considered to pose a higher risk may be denied entry or required to quarantine for a period.

- **How can budgeting provide Covid-19 relief aid to citizens?**

Process of allocating financial resources and providing aid to citizens who have been affected by the COVID-19 pandemic. This can include individuals who have lost their jobs or businesses, or who are experiencing financial hardship because of the pandemic. To budget for this aid, officials would likely analyze data on the economic impact of the pandemic and estimate the financial resources required to aid those in need. This may include direct financial assistance, such as cash payments, or indirect assistance, such as subsidies for rent or utilities.

In conclusion, the expert system can play a critical role in addressing key areas of concern in managing COVID-19. By considering optimal resource allocation, vaccination strategies, continuous versus risk-based measures, risk assessment for travelers, and providing relief aid, we can successfully navigate the challenges posed by the virus. By using an expert system, decision-makers can contribute to evidence-based decision-making and mitigate the impact of the pandemic on individuals, communities, and nations.

5.2 Selection of Approach

Author: Shehab (100%)

Based on the earlier chapter where we discussed various topics including expert systems and machine learning and the problem at hand, moving forward we will focus on Case Based Reasoning also known as CBR for short as an approach to recommend policies, therefore the proposed system will be based on this technique. CBR can be used as a standalone problem solver which retrieves past cases and compares them with a new case, the most similar case is then chosen as the solution. CBR can also be combined with Rule Based Reasoning also known as RBR. RBR is the traditional rule based expert system. According to [41] these two approaches are

fundamentally different from one another in terms of reasoning and either one can be adapted as a recommender system. RBR is an approach to problem solving that relies on rules and these rules are usually represented as IF-THEN conditions to reach a solution. CBR on the other hand, is completely different in that it doesn't rely on rules, but rather historical knowledge as explained earlier, and it works by reusing past solutions to a current problem. It does this by finding the most similar case to a current problem. This historical knowledge is represented as cases which include a problem and a solution property. CBR has the added advantage of continuing learning as cases are solved, they are incrementally added to the case library, but RBR requires modifications of rules which can get complex. According to [42] these two approaches, CBR first RBR second, RBR first CBR second, and both on their own in one system. In our approach, we propose choosing CBR with case adaptation using rules, and include an explanation system which can convert solutions from the systems into human readable format. RBR only or both are out of scope for this research and can be investigated on their own. Lastly, since we are in 2023 and data on Covid-19 is available, CBR becomes an obvious choice to the availability of data which had already historical records of policies have been implemented during Covid-19. The system and its components will be explored and detailed in a later section. However, to set the stage for what is coming, to use CBR we need to consider a similarity metric, in this case for simplicity, we choose Euclidean distance, a case structure, as well a case library which includes a list of cases from the past and their solutions and finally, an adaptation mechanism to improve the recommendations.

5.3 Case Based Reasoning

A case-based reasoning system is proposed to introduce a mechanism for decision makers to recommend policies in the context of Covid-19 as the main pandemic and other similar pandemics based on past (historical) solutions. The system is based on CBR with rule-based adaptation, which solves a new problem by referring to problems in the past and applies the solution used historically for a given problem by a user. This requires building a case library with features that describe each case as a problem with its solution from past experiences. The case library will only consider data from Rhineland-Palatinate state in Germany from 2020 until 2022.

5.3.1 Defining a Case

Author: Shehab (100%)

In this system a case consists of a problem, solution, and an effectiveness score of high, medium, and low, along with other attributes. Here we denote properties that relate to a problem with a prefix “problem_” and the same applies for solution “solution_”. Other attributes include the city name, start and end date and a case ID. A problem is the current situation which the user is trying to find a solution for. In our cases, a problem is the current situation of COVID-19 or any similar pandemic in terms number of active cases, sever active cases, deaths, place, and other demographic attributes as well as vaccination status. A solution on the hand, is several policies namely lockdown, mask, and vaccine. Hence when there is a new problem with the attributes we mentioned, we look for similar problems in the past and apply their solution.

This structure captures a scenario in a before and after fashion. For example, a problem has a solution and an effectiveness score. The score determines how good the solution was when it comes to the impact on reduction of cases, sever cases, and deaths. This way we capture what happened because of a policy and determine its effectiveness. Lastly, each problem and solution in the case library includes a description.

When building the case library, we think of 3 phases of the pandemic.

- Phase one: represents the gradual roll out of policies to control the pandemic.
- Phase two: represents the highest peak of the pandemic and the strictest policies.
- Phase three: represents the gradual withdrawal of policies.

This leads us to build a case library that captures most of the scenarios that can result from a pandemic. In other words, we try to capture the life cycle of a pandemic and build cases based on this idea. Therefore, we suggest building cases for periods of two weeks or more. This way we can capture the effectiveness of policies. Lastly, the case library ideally will include cases from the beginning of 2020 until the end of 2022. This period seems to capture the three phases mentioned earlier. The three phases proposed have been broken to triggering events and waves which will be discussed in the next chapter.

5.3.2 Case Structure

Authors: *Shehab (60%), Sudan (35%), Nishan (5%)*

Here we use key-value pairs to represent the structure of a case. The following is a breakdown of the feature for a case.

General features:

- **Case Id:** denotes the id of a stored case in the case library.
- **Start date:** denote the start period when the problem was captured.
- **End date:** denote the end period when the problem was captured.
- **City:** the place of the problem.

Problem features:

- **Problem description:** the description of a past problem. This is going to be constructed from the rest of the problem features such as population density, population, age distribution, number of active cases at start, number of active cases at end, number of ICU active cases at start, number of ICU active cases at end, number of deaths at start, number of deaths at end, infection rate, mortality rate, and vaccinated population. The system takes these features into account and builds a problem description text based on a predefined template.
- **Population density:** denotes the population density of the city. The classification of population density is based on the information provided by the EC (European Commission) in a report “Testing the degree of urbanization at the global level.” The classification is Urban Center which defines as a city, urban cluster which defines towns and Suburbs and Rural grid cells which define Rural areas and this classification are renamed as High, Medium, and Low respectively for the simplicity of representation of data in our cases.
- **High:** The High population density is defined as at least 1500 inhabitants per sq km and a minimum total population of 50,000.
- **Medium:** Medium density consists of at least 300 inhabitants per sq km and a minimum total population of 5,000.
- **Low:** A low density is considered as below 300 inhabitants per sq km
- **Population:** refers to the number of people in the city.
- **Age distribution:** the median age distribution of the population. when all age values are sorted in either an ascending or descending order, refers to the specific middle age. It is regarded as one of the key indicators in defining an aging population since it can reflect the general changes in the entire population. The system uses the average median age.
- **Number of active cases at start:** denotes the number of active covid cases in the city at the start date.

- **Number of active cases at end:** denotes the number of active covid cases in the city at the end date.
- **Number of ICU active cases at start:** denotes the number of ICU active covid cases in the city at the start date.
- **Number of ICU active cases at end:** denotes the number of ICU active covid cases in the city at the end date.
- **Number of deaths at start:** denotes the total number of deaths at the start date in the city.
- **Number of deaths at end:** denotes the total number of deaths at the end date in the city.
- **Vaccinated population:** refers to the portion of the population who is vaccinated at least one shot. The classification of vaccinated population is determined by the percentage of vaccinated population. The classification provided is based on the labels provided by the WHO (World Health Organization). The classification of the vaccinated population are as follows:
 - **High:** If the vaccinated population is more than or equals to 70% out of the total population of that area then the vaccinated population is considered as high.
 - **Medium:** If the vaccinated population is more than or equals to 40% and less than 70% then it is considered as medium.
 - **Low:** If the vaccinated population is less than 40% then the vaccinated population is classified as low.
- **Infection rate:** refers to the frequency of occurrence in infections. This is stored as float number representing the percentage. The calculation of infection rate will be done using the formula below. Where K will be a constant 100 multiplied to number of infections at that time divided by number of those at risk which will be the population of that area.

$$\text{Rate of infection} = K \times \frac{\text{the number of infections}}{\text{the number of those at risk of infection}}$$

- **Mortality rate:** refers to how severe the virus is. This is stored as float number representing the percentage. The formula represents how many people for every 1000 people in the population died when the pandemic started in Germany.

$$\frac{\text{Total deaths in a given time period}}{\text{Total population in the same time period}} * 1000$$

- **Temperature:** refers to the average temperature between the start and end date period.
- **Humidity:** refers to the average humidity between the start and end date period.

Solution features:

The explanation of each policy level below is according to The Oxford COVID-19 Government Response Tracker (OxCGRT) [43], not paraphrases and quoted directly.

- **Solution description:** the description of past solution for the problem in the case.
- **Lockdown policy:** denotes the type of lockdown policy that was or has been implemented during that period. This is divided into levels:
 - **Level 0:** “No measures”
 - **Level 1:** “Recommend not leaving house.”
 - **Level 2:** “Require not leaving house with exceptions for daily exercise, grocery shopping, and ‘essential’ trips.”
 - **Level 3:** “Require not leaving house with minimal exceptions (e.g., allowed to leave only once every few days, or only one person can leave at a time, etc.)”
- **Mask policy:** denotes the type of mask policy that was or has been implemented during that period. This is divided into levels:
 - **Level 0:** “No measure.”
 - **Level 1:** “Recommended.
 - **Level 2:** “Required in some specified shared/public spaces outside the home with other people present, or some situations when social distancing is not possible.”
 - **Level 3:** “Required in all shared/public spaces outside the home with other people present or all situations when social distancing is not possible.”
 - **Level 4:** “Required outside the home at all times, regardless of location or presence of other people.”
- **Vaccine policy:** denotes the type of vaccine policy that was or has been implemented during that period. This is divided into levels:

- **Level 0:** “No availability”
- **Level 1:** “Availability for ONE of the following: key workers/ clinically vulnerable groups / elderly groups”
- **Level 2:** “Availability for TWO of the following: key workers/ clinically vulnerable groups / elderly groups”
- **Level 3:** “Availability for ALL the following: key workers/ clinically vulnerable groups / elderly groups”
- **Level 4:** “Availability for all three, plus partial additional availability (select broad groups/ages)”
- **Level 5:** “Universal availability.”
- **Effectiveness:** refers to the amount of decrease or increase in cases, sever cases, and infection rate. The effectiveness score is calculated by taking the difference between each feature before and after the period of which the case was captured. For example, we look at the start date of the case and end date of the case and calculate the difference in this period. This is another categorical value which includes the following.
 - **High:** reduced by more than 70%.
 - **Medium:** reduced between 30% to 70%.
 - **Low:** reduced by less than 30%.

5.3.3 Case Solution

Author: Shehab (100%)

A user will enter a problem through a web-based application. The following problem information will be given by the user:

- Start date (required).
- End date.
- City (required).
- Number of active cases at start (required).
- Number of active cases at end.
- Number of ICU active cases at start (required).
- Number of ICU active cases at end.
- Number of deaths at start (required).

- Number of deaths at end.
- Vaccinated population.
- Average temperature during the selected period.
- Average humidity during the selected period.

Other information such as population, population density and infection rate, mortality rate will be determined by the system using the name of the city and number of cases, ICU cases and deaths, respectively.

Once done, the user submits this request to our API which uses CBR to reach a solution. A solution is a description that represents the information received and the solution proposed. Besides the solution description, the system will also fetch the top 3 most similar cases to the current problem, as well as the calculated infection and mortality rate. An example of a solution can be found below:

“In a scenario where the population density is moderate, median age is 44.5, infection rate is 0.18 percentage, and mortality rate is 0.09 percentage, with 12 ICU active cases. It is recommended to implement a level 2 lockdown policy, (require not leaving house with exceptions for daily exercise, grocery shopping, and ‘essential’ trips). A level 2 mask policy (required mask in some specified shared/public spaces outside the home with other people present, or some situations when social distancing not possible). It is important to note that (vaccination available for all the following: key workers/ clinically vulnerable groups / elderly groups). Overall, taking these measures can help control the spread of the disease and minimize the number of severe cases and deaths in the affected area.”

Other similar solutions include:

Case 1 (another case and its solution)

Case 2 (another case and its solution)

Case 3 (another case and its solution)

In this chapter we discussed mainly the different types of questions the expert system can answer, and we selected a question around strategy. We then discussed our approach using CBR and the type of data the CBR needs as well as what the answer from this system will look like. We also looked deeply into each data feature to explain it. In the next chapter, we discuss methodology of

collecting data, collect the data from various sources, discuss different approaches to build case timelines and build these timelines to represent cases.

6. Data Collection and Insights

This section shows the process of collecting data from sources such as Robert Koch Institute and other relevant sources for the case library of the system as well as discussing methodology for data quality, and time frames.

6.1 Data Quality Methodology

Author: Nishan Karki (100%)

Data quality is the king of any kind of data-driven decision-making, the output is mostly dependent on the quality of data being fed rather than the complex models. If the quality of data is bad, the system works as GIGO (Garbage n Garbage Out), so the better the quality, the better the decision-making.

1. Data Profiling:

A meta-data about the data gives the user a high-level overview of the data in use. The number of columns, the datatypes needed for each feature, the number of distinct values in each feature, etc. are some of the information the meta-data provides, making it easier to understand what data are being used and what the data should be.

2. Data Cleaning:

Since the data used is collected from numerous sources, the data in its native format is inconsistent. Some of the common inconsistencies could be the date format, the abbreviation used to represent the city names, such as Koblenz could be represented as KBLZ or KBL depending on the data source, there could be duplication, typo error, wrong data format i.e., a number could be wrongly written as a string such as 4300 or “4300”, removing the outliers that could lead to the misleading decision. All the data must be cleaned and brought into a single uniform state.

3. Data validation:

To add new features to the existing system, the new data must be validated against the data that has already been cleaned and has been considered the standard. So, to feed the new data to the existing system, data reconciliation must be done.

4. Data de-normalization:

The data have their own level of magnitude, and using it will make the system biased i.e., the features which have higher magnitude will be given higher priority and vice versa. The use of categorical values in the model also needs to be encoded into numerical representation and the existing numerical values having higher magnitudes also need to be standardized such that there is no bias. Ex: number of hospitals, it could be a double-digit compared to the number of active or death cases which could go in thousands or even millions.

6.2 Data Collection for Active Cases and Deaths

Author: Sudan (100%)

The COVID-19 data for the state of Rheinland-Pfalz in Germany was collected from the website <https://coronalevel.com/Germany/Rheinland-Pfalz>. This website serves as a valuable resource for accessing up-to-date information on the COVID-19 situation in the region and its source is Robert Koch Institute (RKI) [44]. Daily active cases and total deaths are the data gathered from this website. However, it's crucial to remember that the Robert Koch Institute (RKI), the country's public health organization in charge of illness surveillance and control, is the main source of data for COVID-19 statistics in Germany. To provide accurate and thorough information about the COVID-19 pandemic, the RKI gathers and analyzes data from a variety of sources, including laboratories, hospitals, and health authorities.

```

1 // Link for getting city data
2 selects = [ "Germany/Rheinland-Pfalz/Ahrweiler", "Germany/Rheinland-Pfalz/Altenkirchen-Westerwald", "Germany/Rheinland-Pfalz/Alzey-Worms", "Germany/Rheinland-Pfalz/Bad_D%C3%BCrkheim",
3 "Germany/Rheinland-Pfalz/Bad_Kreuznach", "Germany/Rheinland-Pfalz/Berncastel-Wittlich", "Germany/Rheinland-Pfalz/Birkenfeld", "Germany/Rheinland-Pfalz/Cochem-Zell",
4 "Germany/Rheinland-Pfalz/Donnersbergkreis", "Germany/Rheinland-Pfalz/Eifelkreis_Bitburg-Pr%C3%BCm", "Germany/Rheinland-Pfalz/Frankenthal-Pfalz", "Germany/Rheinland-Pfalz/Germersheim",
5 "Germany/Rheinland-Pfalz/Kaiserslautern_Landkreis", "Germany/Rheinland-Pfalz/Kaiserslautern_Stadt", "Germany/Rheinland-Pfalz/Koblenz", "Germany/Rheinland-Pfalz/Kusel",
6 "Germany/Rheinland-Pfalz/Landau_in_der_Pfalz", "Germany/Rheinland-Pfalz/Ludwigshafen_am_Rhein", "Germany/Rheinland-Pfalz/Mainz", "Germany/Rheinland-Pfalz/Mainz-Bingen",
7 "Germany/Rheinland-Pfalz/Mayen-Koblenz", "Germany/Rheinland-Pfalz/Neustadt_an_der_Weinstra%C3%9Fe", "Germany/Rheinland-Pfalz/Neuwied", "Germany/Rheinland-Pfalz/Pirmasens",
8 "Germany/Rheinland-Pfalz/Rhein-Hunsr%C3%BCck-Kreis", "Germany/Rheinland-Pfalz/Rhein-Lahn-Kreis", "Germany/Rheinland-Pfalz/Rhein-Pfalz-Kreis", "Germany/Rheinland-Pfalz/Speyer",
9 "Germany/Rheinland-Pfalz/S%C3%BCdliche_Weinstra%C3%9Fe", "Germany/Rheinland-Pfalz/S%C3%BCdwestpfalz", "Germany/Rheinland-Pfalz/Trier", "Germany/Rheinland-Pfalz/Trier-Saarburg",
10 "Germany/Rheinland-Pfalz/Vulkaneifel", "Germany/Rheinland-Pfalz/Westerwaldkreis", "Germany/Rheinland-Pfalz/Worms", "Germany/Rheinland-Pfalz/Zweibr%C3%BCcken",
11 ];
12 selects.forEach(function getdata(select) {
13     fetch("https://coronalevel.com/" + select + "/datagroup-total-v3.json") //fetching the json file from the coronalevel website
14     .then((resp) => resp.blob())
15     .then((blob) => {
16         const url = window.URL.createObjectURL(blob);
17         const a = document.createElement("a");
18         a.style.display = "none";
19         a.href = url;
20         // the filename you want
21         filename = select.split("/");
22         a.download = filename[2] + ".json";
23         document.body.appendChild(a);
24         a.click();
25         alert("your file has downloaded for-" + select);
26         window.URL.revokeObjectURL(url);
27     })
28     .catch(() => alert("Failed to download the file"));
29 });

```

Figure 7: Script for downloading data.

Converting JSON to CSV

```

import json
import pandas as pd

import os, glob
path = 'd:/dev/Covid/rhein/'
for filename in glob.glob(os.path.join(path, '*.json')):
    with open(os.path.join(os.getcwd(), filename), 'r', encoding="utf-8") as f:
        data = json.load(f)
        fname = (data["loc"]["label"])

        f.close()
        new_Data = data['series']
        df = pd.DataFrame ()
        df5 = pd.DataFrame ()
        counter = 0
        for j in new_Data:
            for i in j['data']:
                i.append(fname)
                df = pd.DataFrame (j['data'], columns = ['date', j['kind'], 'city'])
                df['date'] = pd.to_datetime(df['date'],unit='ms')
                df['date'] = pd.to_datetime(df['date']).dt.normalize()

                df5 = pd.concat([df5, df], axis=1)
        df2 = df5.T.drop_duplicates().T

        df2.to_csv('d:/dev/Covid/rhein/'+fname+'.csv', index=False, encoding='utf-8')
        print(f"Successfully converted JSON to CSV for : "+fname)

```

Figure 8 Script for converting JSON to CSV

The provided code is a Python script that automates the conversion of COVID-19 data from JSON files to CSV format for the state of Rheinland-Pfalz in Germany. It iterates through all the JSON files in a specified directory and performs the following steps for each file:

1. Reads the JSON data and extracts the location label (fname).
2. Processes the data by creating a panda DataFrame, converting the date format, and appending the location label to each row.
3. Concatenates the processed DataFrame with a master DataFrame (df5).
4. Removes any duplicate rows from the master DataFrame (df2).
5. Exports the final DataFrame to a CSV file with the location label as the filename.

Filling in the missing cases:

For each city in Rheinland-Pfalz, missing data for active cases from 2020-03-03 to 2020-10-21 were filled in using a method based on the computation of active cases percentage. First, for each day within the given date period, the active cases percentage for the entire Rheinland-Pfalz region was calculated. The ratio of active cases to the entire population in Rheinland-Pfalz is represented by this percentage.

The population of each city in Rheinland-Pfalz was then divided by this percentage. An estimated number of active cases for each city was calculated by multiplying the population by the percentage of active cases.

This ensured that each city had a representative number of current cases based on the regional percentage and their respective population by filling in the missing data for the selected period range using the predicted values.

```
import os, glob

cases_path = 'd:/dev/Covid/rhein/final_data'
df = pd.read_csv('Rheinland-pfalz.csv')

df['state_population'] = 4085831
df['percentage'] = (df.active / df.state_population) * 100

for filename in glob.glob(os.path.join(cases_path, '*.csv')):
    with open(os.path.join(os.getcwd(), filename), 'r', encoding="utf-8") as f:
        base = os.path.basename(filename)
        # reading two csv files
        data1 = pd.read_csv(f)
        city_name = data1["city"][0]
        city_details = cities[city_name]["population"]

        # using merge function by setting how='inner'
        output1 = pd.merge(data1, df, on=['date'], how='outer')
        output1['city_population'] = city_details
        output1['predicted_active'] = round((output1.percentage * output1.city_population) / 100)

        output1 = output1.rename({'city_x': 'city', 'deaths_x': 'deaths'}, axis=1)
        output1['city'] = city_name
        output1.loc[output1["active_cases"].isnull(), 'active_cases'] = output1["predicted_active"]

        output2 = pd.merge(output1, converted_dates, on=['date', 'city'], how='inner')
        output2.loc[output2["deaths"].isnull(), 'deaths'] = output2["value"]
        output3 = pd.merge(output2, converted_dates_cases, on=['date', 'city'], how='inner')
        output3.loc[output3["cases"].isnull(), 'cases'] = output3["value1"]

        output3 = output3.sort_values(by='date')
        output3 = output3.drop(columns=['city_y', 'deaths_y', 'date.1', 'recovered', 'active', 'confirmed', 'value1', 'value',
        'predicted_active', 'predicted_active', 'city_population', 'percentage', 'state_population'])
        output3.to_csv('d:/dev/Covid/rhein/predicted/'+os.path.splitext(base)[0]+"_pred.csv", index=False, encoding='utf-8')
        print(output2)
```

Figure 9 Script for filling missing active cases.

The above script was used to calculate the estimated active cases for each city based on the percentage of active cases in Rheinland-Pfalz and population of the respected city.

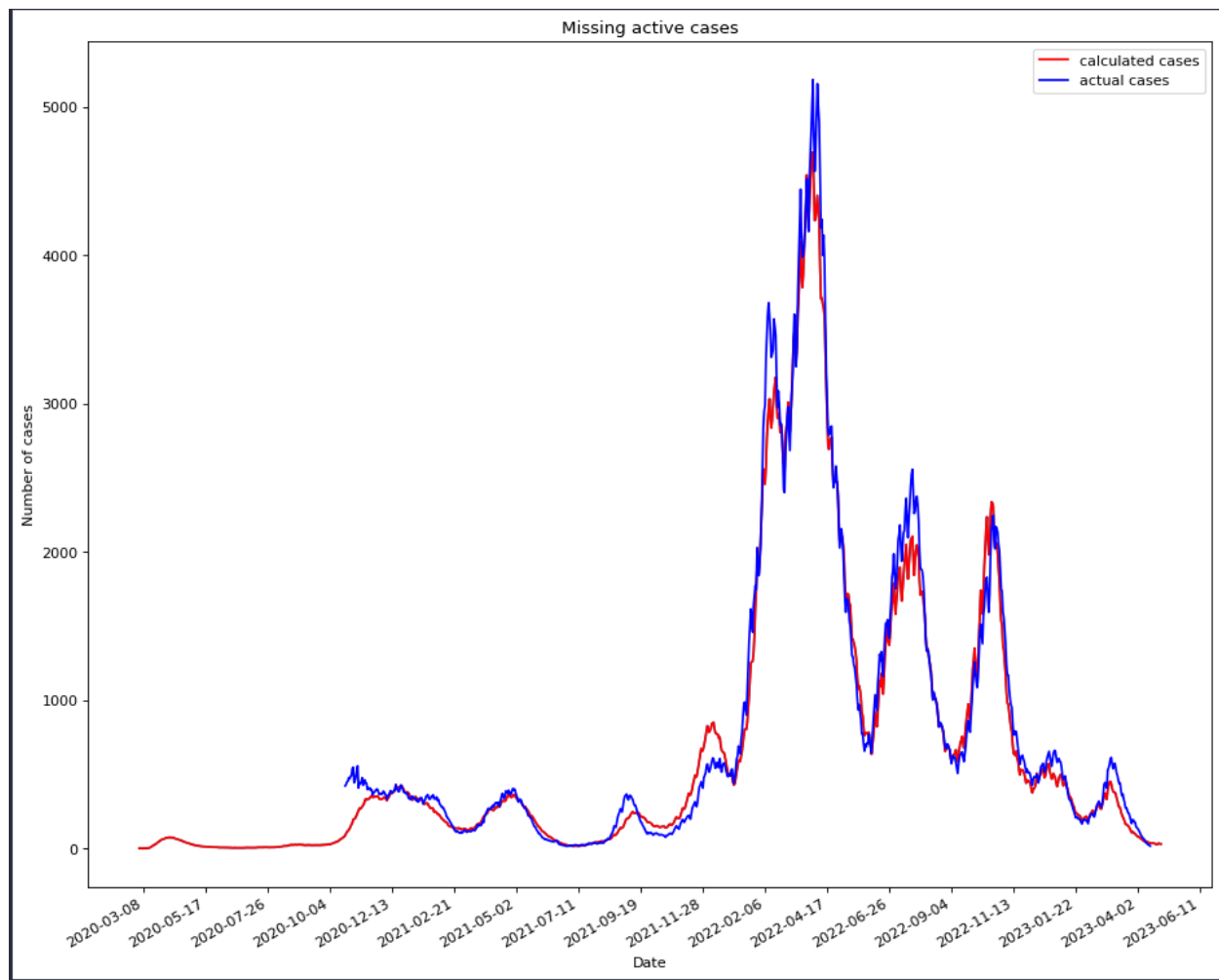


Figure 10 Plot of actual active cases and calculated active cases.

Based on the percentage calculation approach, the graphic compares the estimated active cases to the actual active cases for the city of Koblenz. Dates are represented on the x-axis, and the number of current cases is shown on the y-axis.

The red line on the plot represents the calculated active cases using the percentage calculation method. This line shows the estimated number of infected people in Koblenz based on the city's population and the percentage of infected people in the entire Rhineland-Palatinate region. Estimates of expected number of active cases are represented by red lines.

Actual active cases in Koblenz are depicted by the blue line. This line displays accurate information that was gathered from reputable sources or government records. It displays the actual number of reported active cases for each relevant day.

The disparities between the estimated and actual active instances can be seen by comparing the red and blue lines. The accuracy of the % computation method may be evaluated using this visual representation, and any differences between the estimated and reported statistics can be found. Such a plot aids in comprehending the efficiency of the estimating method and can reveal information about the accuracy of the calculated values in comparison to the actual data.

6.3 Data Collection for Weather Data

Authors: Neil (50%), Rahul (50%)

The Weather data for the state of Rhineland Palatinate was collected with the help of the BrightSky API <https://brightsky.dev>. This API helps fetch weather data like the Average Temperature as well as Average Humidity. BrightSky API obtains this data from the DWD. The DWD (Deutscher Wetterdienst) is the national meteorological service of Germany. It is responsible for monitoring and forecasting weather conditions, issuing weather warnings, and providing climatological information for Germany.

For the collection of this data, we wrote a script to access the BrightSky API to obtain district wise weather data for the state of Rhineland Palatinate for the duration 01-04-2020 to 31-12-2022.

```

In [1]: import requests
        from bs4 import BeautifulSoup
        import json
        import pandas as pd
        from datetime import date, timedelta

        def read_latlong():
            return pd.read_excel('latlong.xlsx')

        def getData(start_date, end_date, latlong_df):
            main_df = []
            for single_date in datarange(start_date, end_date):
                currentdate = single_date.strftime("%Y-%m-%d")
                for (colname, colval) in latlong_df.iterrows():
                    generated_url = generateUrl(colval.values[1], colval.values[2], currentdate)
                    individual_df = writetocsv(generated_url, colval.values[0], currentdate)
                    main_df.append(individual_df)
            return main_df

        def datarange(start_date, end_date):
            for n in range(int((end_date - start_date).days)):
                yield start_date + timedelta(n)

        def generateUrl(latitude, longitude, date):
            return f'https://api.brightsky.dev/weather?lat={latitude}&lon={longitude}&date={date}'

        def writetocsv(url, districtName, currentdate):
            req = requests.get(url)
            soup = BeautifulSoup(req.content, 'html.parser')

            try:
                json_dicti = json.loads(soup.text)
                if 'weather' in json_dicti:
                    tocsv = pd.DataFrame(json_dicti['weather'])
                    tocsv.insert(0, 'District Name', districtName)
                    tocsv.insert(1, 'Current Date', currentdate)
                    return tocsv
                else:
                    print(f"No weather data found for {districtName} on {currentdate}")
                    return pd.DataFrame()

            except json.JSONDecodeError:
                print(f"Failed to parse JSON response for {districtName} on {currentdate}")
                return pd.DataFrame()

            except Exception as e:
                print(f"An error occurred while scraping {districtName} on {currentdate}: {str(e)}")
                return pd.DataFrame()

        def write_to_csv(main_dataframe):
            with open("2022DataUpdated.csv", "a", newline="") as csvfile:
                count = 0
                print("Writing to file initiated")
                for each_df in main_dataframe:
                    if count == 0:
                        each_df.to_csv(csvfile, index=False)
                        count = 1
                    else:
                        each_df.to_csv(csvfile, header=False, index=False)
                print("Writing to file Finished")

```

- The **read_latlong** function reads the latitude and longitude data from an Excel file to load the data into a pandas data frame. The function then returns this data frame containing the latitude and longitude values.
- The **getData** function is responsible for retrieving weather data for a range of dates and geographical coordinates. Within the function, there is a loop that iterates over each date within the specified range using the **daterange** generator function. This nested loop allows processing weather data for each combination of date and geographical coordinates. For each iteration of the nested loop, the **generateUrl** function is called and a URL is constructed to fetch weather data from the BrightSky API for the specific location and date. The writetocsv function is then

called, passing the generated URL, district name, and current date as arguments. This function retrieves the weather data from the BrightSky API, parses the response, and returns a pandas DataFrame containing the weather information. The obtained DataFrame is appended to the `main_df` list. Once all iterations are completed, the `main_df` list contains the weather data for all dates and geographical coordinates which is written to a CSV file using the `write_to_csv` function to store the weather data for analysis or other purposes.

- The **`daterange`** function is a generator function that generates a sequence of dates within the specified range. The range is determined by calculating the number of days between the **`start_date`** and **`end_date`**. The **`yield`** keyword is used to yield the current date as the output of each iteration of the loop.
- The **`generateUrl`** function is called to generate the URL for the BrightSky API request by combining the latitude, longitude, and current date to construct the API endpoint URL. Once the URL is constructed, it is returned as the output of the `generateUrl` function. The generated URL will be used in the `writetocsv` function to make a request to the BrightSky API and retrieve the weather data for the specified location and date.
- The **`writetocsv`** function is responsible for retrieving weather data from the BrightSky API, parsing the response, and converting it into a pandas DataFrame that can be written to a CSV file. The function starts by making an HTTP GET request to the API using the **`requests.get`** function and the `url` parameter. The response from the API is obtained, and the content of the response is accessed using the `content` attribute. The content of the API response is then parsed using BeautifulSoup with the **`'html.parser'`** parser. The function attempts to parse the JSON content of the API response using **`json.loads`** and assigns it to the **`json_dicti`** variable. If the `'weather'` key is present, a pandas DataFrame called `tocsv` is created with the weather data as its input. If the `'weather'` key is not present in the `json_dicti`, it indicates that no weather data was found for the specified district and date. Additionally, two new columns, `'District Name'` and `'Current Date'`, are inserted at the beginning of the DataFrame. Finally, the constructed DataFrame is returned as the output of the function.

6.4 Data Collection of ICU, Death Cases, Severe Cases, Population, and Average Age

Author: Nishan (100%)

The data for ICU, deaths, severe cases, population, and average age has been retrieved from the sites <https://www.corona-daten-deutschland.de/dataset/todesfaelle/resource/e99c8fb7-b8f9-4cc6-801c-debf2b1301a6>, <https://www.corona-daten-deutschland.de/dataset/intensivstationen/resource/da2e682f-6154-434f-bd79-01e816c38ad6>, and <https://zhujiworld.com/de/1580062-birkenfeld/>.

Corona-Daten-Deutschland is a project of the Federal Ministry for Economic Affairs and Climate Protection, which is an interdisciplinary consortium commissioned to collect data related to the crisis in Corona.

The focus for this data collection is set to Rhineland-Pfalz state.

Cleaning ICU data:

1. Renaming columns from German to English:

The data in its native format was presented using German language, so first to get the proper overview of what each column exactly meant, we are converting the column names from German to English.

The below screenshot represents the names of the columns before and after the transformation:

```
1 icu_data.columns
Index(['_id', 'ags2', 'bundesland', 'ags5', 'kreis', 'datum', 'kr_its_kh',
      'kr_its_meldeb', 'kr_its_bett_f', 'kr_its_bett_f_18', 'kr_its_bett_b',
      'kr_its_bett_b_18', 'kr_its_bett', 'kr_its_bett_ew', 'kr_its_inf',
      'kr_its_inf_b', 'kr_its_inf_b_ant'],
      dtype='object')
```

```
1 icu_data.columns
Index(['date', 'city', 'total_intensive_cares', 'free_beds_in_ICU',
      'occupied_beds_in_ICU', 'total_ICU_beds', 'COVID_patients_in_ICU',
      'no_severe_cases'],
      dtype='object')
```

2. Selecting data for Rhineland-Pfalz:

The data contains all the states of Germany, to drill down our analysis and focus only on Rhineland, we are filtering out the data and selecting only Rhineland state.

The below screenshot shows all the available states and the filtering out of the rest and only selecting the required columns for further analysis.

```
1 icu_data['state'].unique()
array(['Schleswig-Holstein', 'Hamburg', 'Niedersachsen', 'Bremen',
      'Nordrhein-Westfalen', 'Hessen', 'Rheinland-Pfalz',
      'Baden-Württemberg', 'Freistaat Bayern', 'Saarland', 'Berlin',
      'Brandenburg', 'Mecklenburg-Vorpommern', 'Sachsen',
      'Sachsen-Anhalt', 'Thüringen'], dtype=object)

1 icu_data = icu_data[icu_data['state'] == 'Rheinland-Pfalz']

1 icu_data.columns
Index(['state', 'city', 'date', 'total_intensive_cares', 'free_beds_in_ICU',
      'occupied_beds_in_ICU', 'COVID_patients_in_ICU', 'no_severe_cases',
      'total_ICU_beds'],
      dtype='object')

1 icu_data = icu_data[['date', 'city', 'total_intensive_cares', 'free_beds_in_ICU',
2                      'occupied_beds_in_ICU', 'total_ICU_beds', 'COVID_patients_in_ICU', 'no_severe_cases']]

1 icu_data.columns
Index(['date', 'city', 'total_intensive_cares', 'free_beds_in_ICU',
      'occupied_beds_in_ICU', 'total_ICU_beds', 'COVID_patients_in_ICU',
      'no_severe_cases'],
      dtype='object')

1 icu_data.head(3)
```

	date	city	total_intensive_cares	free_beds_in_ICU	occupied_beds_in_ICU	total_ICU_beds	COVID_patients_in_ICU	no_severe_cases
0	2020-04-24	Stadt Koblenz	4	45	67	112	9	8
1	2020-04-25	Stadt Koblenz	4	42	70	112	8	7
2	2020-04-26	Stadt Koblenz	4	36	72	108	8	7

Cleaning Death Data:

1. Selecting only Rhineland data and then renaming columns from German to English:

In this step, we will first select only data related to Rhineland, and then do the transformations i.e., translating the column names, selecting the required columns and performing the aggregation.

```
: 1 death_data['bundesland'].unique()
: array(['Schleswig-Holstein', 'Hamburg', 'Niedersachsen', 'Bremen',
      'Nordrhein-Westfalen', 'Hessen', 'Rheinland-Pfalz',
      'Baden-Württemberg', 'Freistaat Bayern', 'Saarland', 'Berlin',
      'Brandenburg', 'Mecklenburg-Vorpommern', 'Sachsen',
      'Sachsen-Anhalt', 'Thüringen'], dtype=object)

: 1 rheinland_pfalz = death_data[death_data['bundesland'] == 'Rheinland-Pfalz']
```

```
1 rheinland_pfalz.columns

Index(['kreis', 'datum', 'kr_tod_md', 'kr_tod_a0004', 'kr_tod_a0514',
      'kr_tod_a1534', 'kr_tod_a3559', 'kr_tod_a6079', 'kr_tod_a80',
      'kr_tod_99', 'kr_tod_m', 'kr_tod_w', 'kr_tod_u', 'kr_sterbe_rate'],
      dtype='object')
```

```
1 new_columns = {
2     "kreis": "city",
3     "datum": "date",
4     "kr_tod_md": "Deaths",
5     "kr_tod_a0004": "age_upto_4",
6     "kr_tod_a0514": "age_between_5_to_14",
7     "kr_tod_a1534": "age_between_15_to_34",
8     "kr_tod_a3559": "age_between_35_to_59",
9     "kr_tod_a6079": "age_between_60_to_79",
10    "kr_tod_a80": "age_above_or_80",
11    "kr_tod_99": "age_unknown",
12    "kr_tod_m": "total_male_death",
13    "kr_tod_w": "total_female_death",
14    "kr_tod_u": "gender_unknown",
15    "kr_sterbe_rate": "cumulative_death_on_confirmed_cases"
16 }
```

```
1 rheinland_pfalz = rheinland_pfalz.rename(columns=new_columns)
```

```
1 rheinland_pfalz.columns

Index(['city', 'date', 'Deaths', 'age_upto_4', 'age_between_5_to_14',
      'age_between_15_to_34', 'age_between_35_to_59', 'age_between_60_to_79',
      'age_above_or_80', 'age_unknown', 'total_male_death',
      'total_female_death', 'gender_unknown',
      'cumulative_death_on_confirmed_cases'],
      dtype='object')
```

From the columns available, we will later aggregate the data and only select the deaths rather than different categories. This step will be implemented in the later stage once all the remaining data is available, and all the data will be merged.

2. Defining the minimum and maximum date threshold:

ICU data is one of the most crucial for analysis, so we are taking out the minimum and the maximum date from this dataset so that we can only get the data from these defined dates to maintain consistency and remove any possible outliers.

```
1 icu_data['date'].min()
```

```
1: '2020-04-24'
```

```
1 icu_data['date'].max()
```

```
1: '2022-12-31'
```

Here, the minimum date is 2020-04-24 and the maximum date is 2022-12-31, now based on these dates, we will only pull the data from the rest of the dataset to maintain consistency in our entire dataset.

```
1 rheinland_pfalz_death_demo = rheinland_pfalz_death[(rheinland_pfalz_death['date'] >= '2020-04-24') & \
2 (rheinland_pfalz_death['date'] <= '2022-12-31')]
```

But still the number of cities doesn't match in the dataset between the ICU and Deaths data, below, we can see there is one more extra city in the death dataset:

```
1 len(icu_data['city'].unique())
```

35

```
1 len(rheinland_pfalz_death_demo['city'].unique())
```

36

3. Dropping the city which is not common in the datasets:

```
1 icu_cities = [city for city in icu_data['city'].unique()]
2 death_cities = [city for city in rheinland_pfalz_death_demo['city'].unique()]
```

```
1 for city in death_cities:
2     if city not in icu_cities:
3         print(city)
```

Rhein-Pfalz-Kreis

```
1 final_df = rheinland_pfalz_death_demo[rheinland_pfalz_death_demo['city'] != 'Rhein-Pfalz-Kreis']
```

```
1 final_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 34370 entries, 147166 to 184407
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   date                  34370 non-null  datetime64[ns]
1   city                  34370 non-null  object
2   Deaths               34370 non-null  int64
3   total_male_death     34370 non-null  int64
4   total_female_death   34370 non-null  int64
5   gender_unknown       34370 non-null  int64
dtypes: datetime64[ns](1), int64(4), object(1)
memory usage: 1.8+ MB
```

```
1 icu_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 34365 entries, 139436 to 173800
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   date                  34365 non-null  object
1   city                  34365 non-null  object
2   total_intensive_cares 34365 non-null  int64
3   free_beds_in_ICU      34365 non-null  int64
4   occupied_beds_in_ICU  34365 non-null  int64
5   total_ICU_beds        34365 non-null  int64
6   COVID_patients_in_ICU 34365 non-null  int64
7   no_severe_cases       34365 non-null  int64
dtypes: int64(6), object(2)
memory usage: 2.4+ MB
```

The number of records is still not the same in the dataset, now we will see which city has the missing data and then handle it:

```
: 1 cols_to_test = icu_data['city'].unique().tolist()
```

```
: 1 def print_record_len(city):
2     print(f"ICU: {len(icu_data[icu_data['city'] == city])} -> {city}")
3     print(f"Death: {len(final_df[final_df['city'] == city])}-> {city}")
4     print("-----")
5
6
7 for city in cols_to_test:
8     print_record_len(city)
```

```

Death: 982-> Stadt Mainz
-----
ICU: 982 -> Stadt Neustadt a.d. W.
Death: 982-> Stadt Neustadt a.d. W.
-----
ICU: 982 -> Stadt Pirmasens
Death: 982-> Stadt Pirmasens
-----
ICU: 982 -> Stadt Speyer
Death: 982-> Stadt Speyer
-----
ICU: 982 -> Stadt Worms
Death: 982-> Stadt Worms
-----
ICU: 982 -> Stadt Zweibrücken
Death: 982-> Stadt Zweibrücken
-----
ICU: 982 -> Alzey-Worms
Death: 982-> Alzey-Worms
-----
ICU: 982 -> Bad Dürkheim
Death: 982-> Bad Dürkheim
-----
ICU: 982 -> Donnersbergkreis
Death: 982-> Donnersbergkreis
-----
ICU: 982 -> Germersheim
Death: 982-> Germersheim
-----
ICU: 982 -> Kaiserslautern
Death: 982-> Kaiserslautern
-----
ICU: 982 -> Kusel
Death: 982-> Kusel
-----
ICU: 982 -> Südliche Weinstraße
Death: 982-> Südliche Weinstraße
-----
ICU: 977 -> Mainz-Bingen
Death: 982-> Mainz-Bingen
-----
ICU: 982 -> Südwestpfalz
Death: 982-> Südwestpfalz
-----

```

The missing record was from Mainz-Bingen city, as the difference between the records is 5 which were the exact missing count. So, the records that were missing for the city were:


```

1 mainz_bingen_icu = icu_data[icu_data['city'] == 'Mainz-Bingen']
2 mainz_bingen_final_df = final_df[final_df['city'] == 'Mainz-Bingen']

1 len(mainz_bingen_icu)

977

1 len(mainz_bingen_final_df)

982

1 mainz_date_cols = mainz_bingen_icu['date'].tolist()

1 mainz_date_cols = [str(date) for date in mainz_date_cols]

1 mainz_bingen_final_df_cols = mainz_bingen_final_df['date'].tolist()

1 mainz_bingen_final_df_cols = [str(date).split(" ")[0] for date in mainz_bingen_final_df_cols]

1 for date in mainz_bingen_final_df_cols:
2     if date not in mainz_date_cols:
3         print(date)

2022-07-30
2022-08-09
2022-08-10
2022-08-11
2022-08-12

```

So, after filtering out these dates in Death data, the data now has equal number of records.

```

1 final_df.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 34365 entries, 147166 to 184407
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   date                   34365 non-null  datetime64[ns]
1   city                   34365 non-null  object
2   Deaths                34365 non-null  int64
3   total_male_death       34365 non-null  int64
4   total_female_death     34365 non-null  int64
5   gender_unknown         34365 non-null  int64
dtypes: datetime64[ns](1), int64(4), object(1)
memory usage: 1.8+ MB

1 icu_data.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 34365 entries, 139436 to 173800
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   date                   34365 non-null  object
1   city                   34365 non-null  object
2   total_intensive_cares  34365 non-null  int64
3   free_beds_in_ICU      34365 non-null  int64
4   occupied_beds_in_ICU  34365 non-null  int64
5   total_ICU_beds        34365 non-null  int64
6   COVID_patients_in_ICU 34365 non-null  int64
7   no_severe_cases       34365 non-null  int64
dtypes: int64(6), object(2)
memory usage: 2.4+ MB

```

4. Merging the data:


```
1 final_df = pd.merge(final_df, icu_data, on=['date', 'city'])

1 final_df.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 34365 entries, 0 to 34364
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                -
0   date                                34365 non-null  datetime64[ns]
1   city                                34365 non-null  object
2   Deaths                             34365 non-null  int64
3   total_male_death                    34365 non-null  int64
4   total_female_death                  34365 non-null  int64
5   gender_unknown                      34365 non-null  int64
6   total_intensive_cares                34365 non-null  int64
7   free_beds_in_ICU                    34365 non-null  int64
8   occupied_beds_in_ICU                34365 non-null  int64
9   total_ICU_beds                      34365 non-null  int64
10  COVID_patients_in_ICU               34365 non-null  int64
11  no_severe_cases                     34365 non-null  int64
dtypes: datetime64[ns](1), int64(10), object(1)
memory usage: 3.4+ MB
```

Now, we will also merge the population and age data and at the end filter out the columns we won't need:

Cleaning Population and Age data:

```
1 population_per_cities = [{"city": cities[i], "population": population[i]} for i in range(0, len(cities))]
```

```
1 population_per_cities
```

```
[{'city': 'Stadt Koblenz', 'population': '114024'},
 {'city': 'Ahrweiler', 'population': '128146'},
 {'city': 'Altenkirchen (Ww)', 'population': '6042'},
 {'city': 'Bad Kreuznach', 'population': '51695'},
 {'city': 'Birkenfeld', 'population': '7075'},
 {'city': 'Cochem-Zell', 'population': '61864'},
 {'city': 'Mayen-Koblenz', 'population': '214259'},
 {'city': 'Neuwied', 'population': '65137'},
 {'city': 'Rhein-Hunsrück-Kreis', 'population': '103026'},
 {'city': 'Rhein-Lahn-Kreis', 'population': '122308'},
 {'city': 'Westerwaldkreis', 'population': '201597'},
 {'city': 'Stadt Trier', 'population': '110636'},
 {'city': 'Bernkastel-Wittlich', 'population': '112262'},
 {'city': 'Eifelkreis Bitburg-Prüm', 'population': '98561'},
 {'city': 'Vulkaneifel', 'population': '60603'},
 {'city': 'Trier-Saarburg', 'population': '148945'},
 {'city': 'Stadt Frankenthal (Pfalz)', 'population': '48561'},
 {'city': 'Stadt Kaiserslautern', 'population': '99845'},
 {'city': 'Stadt Landau in der Pfalz', 'population': '46677'},
 {'city': 'Stadt Ludwigshafen a. Rh.', 'population': '172145'},
 {'city': 'Stadt Mainz', 'population': '217118'},
 {'city': 'Stadt Neustadt a.d. W.', 'population': '53148'},
 {'city': 'Stadt Pirmasens', 'population': '40403'},
 {'city': 'Stadt Speyer', 'population': '50378'},
 {'city': 'Stadt Worms', 'population': '83330'},
 {'city': 'Stadt Zweibrücken', 'population': '34209'},
 {'city': 'Alzey-Worms', 'population': '129244'},
 {'city': 'Bad Dürkheim', 'population': '18790'},
 {'city': 'Donnersbergkreis', 'population': '75101'},
 {'city': 'Germersheim', 'population': '20874'},
 {'city': 'Kaiserslautern', 'population': '99845'},
 {'city': 'Kusel', 'population': '5594'},
 {'city': 'Südliche Weinstraße', 'population': '111279'},
 {'city': 'Mainz-Bingen', 'population': '212420'},
 {'city': 'Südwestpfalz', 'population': '95113'}]
```

```
1 pop_per_city = pd.DataFrame(population_per_cities)
```

Now, we will be filtering out the columns i.e., removing columns of death info that belongs to different groups and categories, we only need the total death column.

```
1 data = pd.read_csv("cleaned_data.csv" , index_col=False)
2 data.head(2)
```

Unnamed: 0	date	city	Deaths	total_male_death	total_female_death	gender_unknown	total_intensive_cares	free_beds_in_ICU	occupied_beds_in_ICU	to
0	0	2020-04-24 Stadt Koblenz	0	0	0	0	4	45	67	
1	1	2020-04-25 Stadt Koblenz	0	0	0	0	4	42	70	

```

1 normal_data = data[['date', 'city', 'Deaths', 'total_intensive_cares', 'free_beds_in_ICU',
2                     'occupied_beds_in_ICU', 'total_ICU_beds',
3                     'COVID_patients_in_ICU', 'no_severe_cases']]

```

```

1 normal_data.head()

```

	date	city	Deaths	total_intensive_cares	free_beds_in_ICU	occupied_beds_in_ICU	total_ICU_beds	COVID_patients_in_ICU	no_severe_cases
0	2020-04-24	Stadt Koblenz	0	4	45	67	112	9	8
1	2020-04-25	Stadt Koblenz	0	4	42	70	112	8	7
2	2020-04-26	Stadt Koblenz	0	4	36	72	108	8	7
3	2020-04-27	Stadt Koblenz	0	4	34	68	102	9	7
4	2020-04-28	Stadt Koblenz	0	4	40	61	101	8	6

1. Merging the population data based on the city names:

```

: 1 normal_data_info = (normal_data.merge(pop_per_city, left_on='city',
2                                       right_on='city'))

```

```

: 1 normal_data_info.head(2)

```

	index	date	city	Deaths	total_intensive_cares	free_beds_in_ICU	occupied_beds_in_ICU	total_ICU_beds	COVID_patients_in_ICU	no_severe_cases	pop
0	0	2020-04-24	Stadt Koblenz	0	4	45	67	112	9	8	
1	1	2020-04-25	Stadt Koblenz	0	4	42	70	112	8	7	

```

: 1 normal_data_info.columns

```

```

: Index(['index', 'date', 'city', 'Deaths', 'total_intensive_cares',
'free_beds_in_ICU', 'occupied_beds_in_ICU', 'total_ICU_beds',
'COVID_patients_in_ICU', 'no_severe_cases', 'population'],
      dtype='object')

```

2. Concatenating the average age with the respective cities:

```
: 1 ages = ["43.3", "46", "44.5", "46", "46", "46.42", "44.63", "44.53", "45.27", "45.44",
2         "44.37", "41.50", "45.26", "44.03", "46.34", "44.28", "44.19", "42.64", "42.52", "41.70",
3         "41.07", "45.75", "46.04", "44.84", "43", "44.96", "43.95", "46.33", "44.97",
4         "43.71", "44.23", "46.07", "45.79", "44.32", "47.21"]
```

```
: 1 cities
```

```
: ['Stadt Koblenz',
'Ahrweiler',
'Altenkirchen (WV)',
'Bad Kreuznach',
'Birkenfeld',
'Cochem-Zell',
'Mayen-Koblenz',
'Neuwied',
'Rhein-Hunsrück-Kreis',
'Rhein-Lahn-Kreis',
'Westerwaldkreis',
'Stadt Trier',
'Bernkastel-Wittlich',
'Eifelkreis Bitburg-Prüm',
'Vulkaneifel',
'Trier-Saarburg',
'Stadt Frankenthal (Pfalz)',
'Stadt Kaiserslautern',
'Stadt Landau in der Pfalz',
'Stadt Ludwigshafen a. Rh.',
'Stadt Mainz',
'Stadt Neustadt a.d. W.',
'Stadt Pirmasens',
'Stadt Speyer',
'Stadt Worms',
'Stadt Zweibrücken',
'Alzey-Worms',
'Bad Dürkheim',
'Donnnersbergkreis',
'Germersheim',
'Kaiserslautern',
'Kusel',
'Südliche Weinstraße',
'Mainz-Bingen',
'Südwestpfalz']
```

```
: 1 avg_cities = [{"city": cities[i], "avg_age": ages[i]} for i in range(0, len(ages))]
```

```
: 1 avg_cities_df = pd.DataFrame.from_dict(avg_cities)
```

3. Merging the age data with the ICU and population data with respect to the city names:

```

1 normal_data_per_day = (normal_data_info.merge(avg_cities_df, left_on='city',
2                                           right_on='city'))

```

```

1 normal_data_per_day.head()

```

	index	date	city	Deaths	total_intensive_cares	free_beds_in_ICU	occupied_beds_in_ICU	total_ICU_beds	COVID_patients_in_ICU	no_severe_cases	pop
0	0	2020-04-24	Stadt Koblenz	0	4	45	67	112	9	8	
1	1	2020-04-25	Stadt Koblenz	0	4	42	70	112	8	7	
2	2	2020-04-26	Stadt Koblenz	0	4	36	72	108	8	7	
3	3	2020-04-27	Stadt Koblenz	0	4	34	68	102	9	7	
4	4	2020-04-28	Stadt Koblenz	0	4	40	61	101	8	6	

```

1 normal_data_per_day.columns

```

```

Index(['index', 'date', 'city', 'Deaths', 'total_intensive_cares',
      'free_beds_in_ICU', 'occupied_beds_in_ICU', 'total_ICU_beds',
      'COVID_patients_in_ICU', 'no_severe_cases', 'population', 'avg_age'],
      dtype='object')

```

6.5 Cases Time Frames Collection Methodology

Author: Christian Asnel (50%), Prajwal (50%)

Creating clear and specific time frames for building cases is essential to evaluate the effectiveness of policies during their implementation. However, this can be a challenging task, particularly in the context of the pandemic, where several events have led to changes in policies.

Building clear and specific time frames for cases requires careful consideration of different approaches. The choice of approach depends on the research question, the availability of data, and the duration of time required to obtain reliable observations. By carefully selecting the appropriate time frame, we can better understand the impact of policies and make informed decisions for future policy implementation.

To establish time frames, four main options were explored:

6.5.1 Option 1: Observing Changes in The Stringency Index

The Stringency Index serves as a gauge to measure the intensity of government responses across a diverse spectrum of policies, with higher values signifying a more rigorous implementation. This index enables us to chronicle shifts in policies and pinpoint events that instigated such changes. It serves as an excellent barometer for observing when changes in policy come into effect, facilitating

the construction of case studies anchored in the Stringency Index. To further enhance our understanding of the context of these policy alterations, we also collate data regarding the dates of these instigating events when such information is accessible.

To illustrate, consider a scenario where a government opts for a nationwide lockdown while maintaining all other policy interventions at the same level of enforcement. A subsequent rise in the Stringency Index would reveal that the policies have become more stringent. By scrutinizing the index's values before and after the lockdown, we can assess the lockdown's efficacy in curbing virus transmission.

However, it is worth noting that the approach has its limitations. Certain policy alterations might not be accurately mirrored in the Stringency Index, and pinpointing the exact triggers for a policy Using this approach, we get 197 cases between 2020 and 2022 out of which only 21 cases are more than 10 days (about 1 and a half weeks) long.

6.5.2 Option 2: Examining Changes in Specific Policy Levels

An alternative method concentrates on policy shifts. This tactic adheres to the logic of the previously discussed approach but hones in on discrete policy amendments. For instance, should a government mandate mask-wearing in public domains, we can monitor this policy's execution and develop case studies accordingly. The quantity of policies under scrutiny directly influences the precision of this approach.

This technique provides a granular perspective on policy shifts and aids in identifying the most effective measures. Nevertheless, challenges may arise when attempting to compare disparate policies as they may operate within distinct temporal parameters, and not every policy might pertain to the research query at hand.

6.5.3 Option 3: Using a Fixed Time Window

This approach involves fixing a specific time frame for building cases. For example, we may decide to analyze the impact of policies implemented during the first six months of the pandemic. Nonetheless, by employing a fixed temporal window, there exists the potential risk of overlooking key triggering events that materialize outside the assigned duration. While this method proves valuable for comparing the efficacy of policies enforced within a given period, it may not offer a comprehensive view of the entire policy environment.

6.5.4 Option 4: Collecting Important Dates

This approach is a mixture of all the previously mentioned approaches. A minimum duration of 10 to 14 days is specified for every time frame: giving a minimum window for observation. Time frames are built by looking at noticeable changes in policies, stringency index and also by referring to the news. Although not completely systematic, we decided to move on with this approach as it provides some of the benefits of all the previously mentioned approaches.

6.5.5 Cases Time Frames

Author: Christian Asnel (50%), Prajwal (50%)

Start Date	End Date	Description
22/1/ 2020	15/2/2020	As of January 27th, there have been a total of 16 reported cases of Covid-19. At this time, there are no mask policies in place. Fortunately, there have been no reported deaths or recoveries thus far. To combat the spread of the virus, comprehensive contact tracing measures have been implemented, ensuring that all identified cases are thoroughly traced and monitored. Public officials have also launched public information campaigns to raise awareness and urge caution regarding Covid-19. Regarding testing, only individuals who exhibit symptoms and meet specific criteria such as being key workers, hospitalized, having contact with a known case, or returning from overseas are being tested. This targeted approach aims to prioritize testing for those most likely to be infected
16/2/2020	15/3/2020	The total number of reported cases currently stands at 4,600. However, there are no mask policies in place to mitigate the spread of the virus. Unfortunately, there have been 9 deaths

		reported thus far, while 22 individuals have successfully recovered from the infection. In terms of policies, there is a requirement to close schools, although it may apply only to certain levels or categories such as high schools or public schools. Additionally, international travel controls are being implemented with screening procedures in place for incoming travellers. To further prevent the spread of the virus, it is recommended to cancel public events as a precautionary measure. People are also advised to stay at home and avoid leaving their houses unless necessary. Furthermore, restrictions on gatherings are being enforced, particularly for very large gatherings, with a limit set above 1000 people to minimize the risk of transmission.
16/3/2020	31/3/2020	The reported number of Covid-19 cases reached 71,808. During this time, there were no mask policies in effect. Tragically, there were 645 deaths reported, but there were also 6,595 recoveries. In terms of policies, on March 18th, all levels of schools were required to close as a preventive measure. Additionally, recommendations were made to discourage internal movement, advising against traveling between regions or cities. International travel controls were implemented, either in the form of a ban on all regions or a complete closure of borders. Workplace closing measures were not enforced during this period. However, public events were required to be cancelled. Furthermore, there were restrictions placed on gatherings, allowing no more than 10 people to congregate.

		Stay-at-home requirements were put in place, mandating individuals to remain in their houses, with exceptions for daily exercise, grocery shopping, and 'essential' trips. Additionally, starting from March 22nd, certain sectors were required to adopt a work-from-home policy, allowing employees to perform their tasks remotely to reduce the risk of virus transmission.
1/4/2020	15/4/2020	During this period, the total number of Covid-19 cases surged to 132,210. Mask policies were implemented, requiring the use of masks to help mitigate the spread of the virus. Unfortunately, there was a significant spike in the number of deaths, with 3,508 reported fatalities. However, there were also 8,541 recoveries. In terms of policies, there were no specific measures implemented to close public transport. However, international travel controls were put in place, either in the form of a ban on all regions or a total closure of borders. All levels of schools were required to close as a preventive measure to limit the transmission of the virus. Additionally, there were restrictions on gatherings, limiting them to 10 people or less to reduce the risk of spreading the virus in large gatherings.
16/4/2020	15/5/2020	At this time period, the total number of reported Covid-19 cases reached 174,975. Mask policies were in effect, mandating the use of masks to help prevent the spread of the virus. Unfortunately, there were 8,028 deaths reported, but there were also 35,196 recoveries.

		Notably, there was a significant decrease in the number of newly infected individuals towards the end of the time period, indicating a positive trend in controlling the spread of the virus. In terms of policies, there were restrictions placed on gatherings, limiting them to 10 people or less to minimize the risk of transmission. Stay-at-home requirements were also implemented, mandating individuals to remain in their houses, with exceptions for daily exercise, grocery shopping, and 'essential' trips. Internal movement restrictions were put in place, aiming to limit movement between regions or cities. However, no specific measures were taken to close public transport. Regarding testing, there was an open public testing policy, which made testing available to asymptomatic individuals, including the option of "drive-through" testing. This approach aimed to identify cases promptly and mitigate the further spread of the virus. Additionally, international travel controls were implemented, either through a ban on all regions or a total closure of borders, to limit the entry and potential importation of the virus from other countries.
16/5/2020	1/6/2020	The total number of reported Covid-19 cases during this period rose to 183,765. Mask policies remained in effect, requiring the use of masks to reduce the transmission of the virus. Encouragingly, there was a further decrease in the number of newly infected individuals, indicating progress in containing the spread of the virus. Additionally, the number of deaths decreased as well, with 8,645 reported

		fatalities, while 6,900 individuals recovered from the infection. In terms of policies, international travel controls were implemented, specifically banning arrivals from certain regions. This measure aimed to limit the introduction of new cases from areas with higher infection rates. Restrictions on gatherings continued, with limitations set at 10 people or less, in order to minimize the risk of transmission in larger groups. Furthermore, internal movement restrictions were in place, aiming to limit the movement of individuals between different regions or cities. As for schools, the policy required the closure of only certain levels or categories, such as high schools or public schools, while others may have remained open. This targeted approach aimed to balance the need for education with the safety of students and staff in specific settings.
2/6/2020	30/6/2020	Throughout the months of June, July, August, and September, the policies remained relatively unchanged due to the stable situation. The policies implemented during this period included the closure of all levels of schools as a preventive measure. Restrictions on gatherings were consistently in place, limiting gatherings to 10 people or less to reduce the risk of transmission. Contact tracing measures were in effect, but limited in scope, not conducted for all cases. Workplace closing was required for some sectors, with the option of working from home provided where applicable. International travel controls continued with bans on arrivals from certain regions, aiming to prevent the introduction of new cases.

		Internal movement restrictions were consistently implemented, aiming to limit movement between regions or cities to contain the spread of the virus. However, there were no specific measures regarding stay-at-home requirements during this time period. Furthermore, a smartphone-based tracing and warning system was activated. This system was designed to alert users when they have come into contact with someone who has tested positive for COVID-19, facilitating early awareness and necessary precautions.
1/7/2020	30/7/2020	The total number of reported Covid-19 cases increased to 209,653 during this period. Mask policies remained in effect, requiring the use of masks to reduce the transmission of the virus. Unfortunately, there were 9,251 deaths reported, while 3,900 individuals recovered from the infection. The number of newly infected individuals was 6,005. Although there was no significant spike in the number of cases for almost two months, there were slight easing of restrictions on gatherings. However, public events were still cancelled as a precautionary measure. The policies included a ban on arrivals from certain regions as part of international travel controls. Regarding schools, it was recommended to close them, potentially as a measure to minimize potential transmission among students and staff. Workplace closing was required during this period, reflecting efforts to reduce the risk of transmission in work settings.

		Restrictions on gatherings were modified to allow gatherings between 11 and 100 people, showing a relaxation in previous limitations. However, internal movement restrictions were still in place to control the spread of the virus. Public events continued to be cancelled, as it was deemed necessary despite the relatively stable situation. Open public testing policy was implemented to facilitate widespread testing for early detection and control of the virus.
1/8/2020	31/8/2020	The total number of reported Covid-19 cases rose to 244,792 during this period. Mask policies remained in effect, requiring the use of masks to reduce the transmission of the virus. Unfortunately, there were 9,389 deaths reported, while 14,980 individuals recovered from the infection. The number of newly infected individuals reached 16,453. In terms of policies, it was recommended to close schools as a precautionary measure, potentially to minimize the spread of the virus among students and staff. Restrictions on gatherings were implemented, allowing gatherings to take place between 11 and 100 people, thereby balancing the need for social interactions with the importance of reducing the risk of transmission. Internal movement restrictions were in place, aiming to control the movement of individuals between different regions or cities as a means to contain the spread of the virus and limit its transmission.

		These policies were put into effect to mitigate the impact of the virus and protect public health by reducing the opportunities for the virus to spread through schools, gatherings, and internal movement.
1/9/2020	30/9/2020	The total number of reported Covid-19 cases has reached 292,911. Mask policies remain in effect, mandating the use of masks to reduce the transmission of the virus. Unfortunately, there have been 9,598 deaths reported, while 21,950 individuals have successfully recovered from the infection. The number of newly infected individuals stands at 26,158. Under the current policies, schools are allowed to remain open, subject to certain conditions that are not specified. Workplace closing is required, indicating that businesses and organizations are mandated to temporarily shut down. Restrictions on gatherings are in place, limiting the number of people to between 11 and 100 for any given gathering. This measure aims to control the potential spread of the virus in larger group settings. Internal movement restrictions are also implemented, meaning that there are limitations on the movement of individuals between different regions or areas to help contain the spread of the virus. Furthermore, public events are required to be cancelled, as a preventive measure to discourage large gatherings and reduce the risk of transmission. These policies are aimed at safeguarding public health by minimizing opportunities for the virus to spread in schools,

		workplaces, gatherings, and during internal movements, while also mitigating the risks associated with public events.
1/10/2020	30/10/2020	During this time frame, there was a notable increase in daily Covid-19 cases, with the numbers ranging from 1,000 to 14,000, reaching the maximum towards the end of the period. These increasing case numbers raised concerns about the need for a strict lockdown. Despite the concerns, schools remained open during this time, implying that education institutions continued to operate under the given policies. However, workplace closing was required, indicating the temporary closure of businesses and organizations. Restrictions on gatherings remained in place, allowing gatherings with a limited number of people ranging from 11 to 100. This measure aimed to balance the need for social interactions with the imperative to reduce the risk of virus transmission. Internal movement restrictions were implemented, meaning that limitations were placed on the movement of individuals between different regions or areas, with the intention of containing the spread of the virus. Furthermore, public events were required to be cancelled as a precautionary measure, discouraging large gatherings and reducing the potential for transmission. These policies were implemented to address the increasing number of cases and protect public health, balancing the need for education, maintaining essential services, and

		reducing the risk of virus transmission in gatherings and movements.
1/11/2020	13/11/2020	The total number of reported Covid-19 cases has risen to 772,822. Mask policies are still in effect, requiring the use of masks to reduce the transmission of the virus. There have been 12,539 reported deaths, but the number of recoveries has increased to 87,000, suggesting that individuals may have developed resistance against the virus or have successfully overcome the infection. The daily infection rate has risen to around 30,000 cases per day, indicating a significant increase in new infections. This rise in cases has prompted the introduction of a "lockdown light" policy, which is a less strict form of lockdown compared to previous measures. Under the new policies, meetings in public are limited to two households and a maximum of 10 people. This restriction aims to reduce the potential for transmission in social gatherings. Additionally, many businesses in the catering, hospitality, and tourism sectors are required to close down once again, in an effort to minimize the spread of the virus in high-risk settings. Despite these measures, schools remain open, suggesting that education institutions continue to operate while adhering to relevant safety guidelines. These policies are implemented to address the increasing number of cases, protect public health, and mitigate the impact on various sectors, while keeping educational institutions functional under specific safety measures.

14/11/2020	30/11/2020	During this period, the total number of reported Covid-19 cases has reached 1,069,763. Mask policies are still in effect, requiring the use of masks to reduce the transmission of the virus. Unfortunately, there have been 16,914 deaths reported, with a significant rise in daily deaths averaging around 200. However, there have also been 103,500 recoveries, indicating ongoing efforts in managing the disease. Developments in vaccines have been initiated in many countries, including Germany, providing hope for potential solutions to combat the virus. The policy of a light lockdown was continued during this period, suggesting that certain restrictions and measures were in place to limit the spread of the virus while allowing some degree of societal and economic activity. As a result, businesses in sectors such as catering, hospitality, and tourism were required to close, acknowledging the heightened risk associated with these industries. Contact tracing was limited, indicating that contact tracing efforts were not carried out for all cases, potentially due to the high number of infections or resource constraints. Additionally, workplace closing was required for some sectors, either through full closure or implementation of work-from-home arrangements, in an effort to minimize the risk of virus transmission in work settings. These policies were implemented as part of ongoing efforts to manage the pandemic, control the spread of the virus, and protect public health. The development of vaccines offers

		hope for a potential long-term solution, while the continuation of a light lockdown and closure of high-risk businesses aim to mitigate the impact of the virus on society.
1/12/2020	31/12/2020	As of the latest data, the total number of reported Covid-19 cases has risen to 1,745,518. Mask policies are still in effect, requiring the use of masks to reduce the transmission of the virus. Unfortunately, the number of deaths has reached 34,313, indicating a significant increase in mortality rates. Additionally, there have been 249,944 newly infected individuals, showcasing the faster spread of the virus during this period. On a positive note, there have been 307,200 recoveries reported. December, in particular, has recorded the highest infection rate since the start of 2020, signifying a concerning acceleration of virus transmission. The severity of the situation is reflected in the highest death rate observed during this period. Discussions were held regarding the implementation of stricter lockdown policies, considering the alarming spread of the virus. However, there were also protests against the Covid-19 restrictions, reflecting the diverse opinions and challenges in enforcing stringent measures. Despite these discussions, a light lockdown was continued, suggesting that a balance between health and economic considerations was maintained. Certain businesses, including those in the catering, hospitality, and tourism sectors, were still required to close due to the increased risk of virus transmission in these settings.

		Contact tracing efforts remained limited, indicating that contact tracing was not conducted for all cases, possibly due to the high volume of infections or logistical constraints. Additionally, workplace closing was required for some sectors, with either full closure or the implementation of work-from-home arrangements, aiming to minimize the risk of virus transmission in work environments. These policies reflect the ongoing efforts to manage the pandemic, adapt to the evolving situation, and mitigate the impact on public health and the economy. The discussions surrounding stricter measures and the protests against restrictions highlight the complexities involved in striking a balance between safeguarding public health and addressing societal concerns.
04/03/2021	18/03/2021	Chancellor Angela Merkel and state leaders in Germany have announced an extension of the current lockdown measures, primarily aimed at controlling the COVID-19 pandemic, until March 28. However, they also highlighted that numerous opportunities for easing restrictions may arise, contingent on the prevailing infection rates. This decision was disclosed late on a Wednesday evening in Berlin. [45]. 11.03.2021: Beginning of third wave. [45]. 12.03.2021: Johnson Johnson vaccine approve [45].
18/03/2021	06.04.2021	RKI: Share of the more dangerous Corona mutant already at 72 percent 23.03.2021: Radical lockdown over Easter

		In response to a significant surge in COVID-19 cases, Germany's national and regional governments have decided to impose the strictest lockdown since the onset of the pandemic one year ago. This lockdown, which is set to affect public, private, and economic activities, is scheduled from April 1st to 5th, encompassing Maundy Thursday through to Easter Monday. The objective of this measure is to curb the third wave of the pandemic. The decision was reached following an extensive meeting that went on for over eleven hours, led by Chancellor Angela Merkel and the premiers of various states on the previous Tuesday night [45].
06/04/2021	23.04.2021	The second phase of Germany's COVID-19 vaccination campaign has been launched with around 35,000 general practitioners nationwide joining the initiative. While some practices have already commenced vaccinations as of this week, others are awaiting their vaccine supply and will begin their vaccination programs in the coming days. Up until now, vaccinations have predominantly been administered in approximately 430 dedicated vaccination centers across the country since the campaign's inception at the end of December. [45]. In a recent update dated April 9, 2021, there's been a noticeable increase in COVID-19 vaccinations in Germany following the involvement of general practitioners. As per the daily vaccination data provided by the Robert Koch Institute, around 656,000 doses were administered on Wednesday, marking an increase of 290,000 from the previous day. The majority of the 35,000 participating general practices across federal states started vaccinations

		on Wednesday. Prior to this surge, the highest number of COVID-19 vaccine doses administered in Germany in a single day had not exceeded 367,000. [45]. 11.04.2021: More than 15 percent of Germans have received first vaccination [45]. 17.04.2021: Merkel vaccinated with Astrazeneca [45].
23/04/2021	07/05/2021	23.04.2021: Federal emergency brake comes into force Citizens of Germany are bracing for new COVID-19 restrictions set to take effect this weekend as a part of the newly introduced Infection Protection Act. The 'emergency brake' provision of this law is activated when the seven-day incidence rate of new infections per 100,000 people in any city or district surpasses 100 for three consecutive days. Upon activation, the rule stipulates that residents should not leave their homes after 10 p.m., with the exception of walking or jogging which is permitted until midnight. Social interaction rules permit only one household to meet with an individual outside their household, excluding children up to the age of 14. Retail shops are allowed to serve customers who can provide a negative COVID-19 test result and have pre-booked an appointment. Classroom attendance at schools will be halted if the incidence rate reaches 165. [45].
07/05/2021	01/06/2021	07.05.2021: Corona rules relaxed for vaccinated and recovered people. [45]. 08.05.2021: Germany allows Astra-Zeneca's COVID-19 vaccine for all adults [45].

01/06/2021	29/06/2021	01.06.2021 Merkel: Federal emergency brake can expire. According to Chancellor Angela Merkel, the "federal emergency brake", a series of standardized measures put in place in response to high COVID-19 case numbers, can cease to be effective by the end of June. Speaking in Berlin, Merkel suggested the possibility of lifting these restrictions. However, she added that if the nationwide infection rate were to rise again, possibly due to new variants of the virus, the "federal emergency brake" could be reactivated swiftly. These standardized measures, established in the Infection Protection Act in April, were designed to remain in effect until June 30 at the latest. [45]. 08.06.2021: Almost half vaccinated. The Robert Koch Institute (RKI) reported on Monday that 38 million individuals, accounting for 45.7% of the population, have received at least one dose of the COVID-19 vaccine. Furthermore, full vaccination has been administered to 17.7 million people, representing 21.3% of the population. Cumulatively, Germany has dispensed a total of 54.9 million doses of the coronavirus vaccine. [45]
29/06/2021	02/08/2021	29.06.2021 RKI: Every second new infection with the delta variant The President of the Robert Koch Institute (RKI), Lothar Wieler, announced in a joint statement with the federal and state health ministers that the highly transmissible delta variant is increasingly prevalent in Germany. Current data suggest that this variant is present in at least 35% of the

		examined samples. However, given that this information is a few days old, it's projected that the real proportion might be closer to 50%. [45].
02/08/2021	20/08/2021	02.08.2021 Relaxations of Corona restrictions [45].
20/08/2021	07/09/2021	20.08.2021 RKI: Fourth wave has begun The Robert Koch Institute (RKI) reports that Germany has entered the fourth wave of the Corona pandemic. In their latest weekly update, published on Thursday evening, it was disclosed that the percentage of positive PCR tests out of about half a million samples from nearly 200 labs has increased from four to six percent within a week until mid-August. Infections are primarily occurring among younger individuals, with some cases traced back to vacation destinations such as the Balkans, Turkey, and Spain. Unvaccinated individuals or those who've only received one dose of the vaccine are at high risk, according to the RKI. The risk for those fully vaccinated is assessed to be moderate. [45].
07/09/2021	29/09/2021	07.09.2021 Majority in favour of stricter Corona rules for the unvaccinated According to a survey conducted by the YouGov opinion research institute, the prospect of imposing stricter Corona restrictions on the unvaccinated compared to the vaccinated, especially concerning access to indoor events, garners majority support. Fifty-eight percent of the survey respondents favored stricter rules for those unvaccinated, while 28 percent thought the same rules should apply to everyone, whether they're vaccinated, recovered, or not.

		Additionally, nine percent of those surveyed voiced their opposition to any measures designed to combat the Corona pandemic. As we approach autumn and winter, the so-called 3G rule is in effect nationwide for specific indoor areas like events and restaurants, which only allows access to those who can provide proof of vaccination, recovery, or a negative test. However, a 2G rule is also under consideration, permitting access solely for those vaccinated or recovered. [45].
29/09/2021	23/10/2021	29.09.2021 Almost three quarters of adults in Germany fully vaccinated As reported by the Robert Koch Institute (RKI), almost three-fourths of the adult population and a third of children aged 12 to 17 in Germany are now fully vaccinated against Corona. The data, which was published on Tuesday, indicates that 74.7% of people aged 18 and above have received the typically necessary second shot. In the 12 to 17 age group, the fully vaccinated percentage stands at 33.2%. The statistics also show that 78.6% of adults and 41% of 12 to 17-year-olds have had at least one initial vaccination. [45].
23/10/2021	04/11/2021	23.10.2021 Corona incidence in triple digits for the first time since May The Robert Koch Institute (RKI) reports that for the first time since the middle of May, the seven-day incidence of COVID-19 in Germany has hit 100. This metric denotes the number of new infections per 100,000 residents over a week's duration. As of Saturday morning, the incidence figure was exactly 100.0, an increase from 70.8 a week

		prior. The rate of COVID-19 patients being hospitalized per 100,000 residents stands at 2.68, which is significantly lower than the peak value of 15.5 recorded at the end of December 2020. [45].
04/11/2021	19/11/2021	04.11.2021 Record high in new corona infections The daily number of new COVID-19 infections in Germany has surged to an all-time high. According to the Robert Koch Institute (RKI), health departments reported a total of 33,949 new COVID-19 cases within a 24-hour period. This number, which was disclosed on Thursday, signifies an increase from the 28,037 infections reported the previous week. The last record high was 33,777 cases on December 18, 2020. Currently, the seven-day nationwide incidence, indicating the number of new infections per 100,000 residents within a week, stands at 154.5. [45].
19/11/2021	13/12/2021	19.11.2021: Demand for drastic Corona measures Lothar Wieler, the head of the Robert Koch Institute, has urged for the implementation of more stringent measures in response to the escalating COVID-19 situation. He suggested that an "emergency brake" must be applied immediately. This includes significantly reducing human contact to curb virus transmission. Wieler recommended people to stay home when feasible, cancel large gatherings, limit attendees at smaller functions, and close high-risk places like poorly ventilated bars and clubs. He warned that the whole of Germany is experiencing a widespread outbreak, declaring it a national emergency. [45].

		Omicron variant appearance [45].
13/12/2021	28/12/2021	Start of Corona vaccinations for children Germany is set to commence COVID-19 vaccinations for children aged five to eleven years old this week. The Health Ministry has announced that over 2.2 million doses of the child specific Biontech/Pfizer vaccine will be distributed to pharmaceutical distributors. Additionally, specific quotas for the vaccine have been allocated to different regions. Medical practices obtain the vaccines via pharmacies. The Permanent Vaccination Commission has recommended vaccination for children aged between five and eleven who are either at high risk of severe COVID-19 or live with high-risk individuals. Parents of healthy children can also opt for vaccination following individual consultation. [45].
28/12/2021	19/01/2022	Stricter contact restrictions nationwide Starting from Tuesday, Germany will uniformly implement the contact restrictions recently passed by federal and state governments. Gatherings will be limited to a maximum of ten individuals, who are either vaccinated or have recovered from the virus, both indoors and outdoors. Unvaccinated individuals are restricted to meeting with two people outside their household. Nightclubs and discos are to be shut down. The 2G or 2G-plus rule, which mandates vaccination or recovery status for access to recreational, cultural, and retail facilities, will remain in effect. However, stores catering to daily needs are exempt. Some regions have adopted stricter regulations, such as restaurant curfews. [45].

19/01/2022	28/01/2022	19.01.2022: Over 100,000 new infections In comparison to other EU nations, Germany is now witnessing a surge in COVID-19 cases that is unparalleled since the start of the pandemic. The Robert Koch Institute (RKI) recorded more than 100,000 daily infections for the first time on Wednesday. In light of this situation, Federal Health Minister Lauterbach emphasized the potential need for a general vaccination mandate, which could be enforced as early as April following a decision by the Bundestag. He reasoned that the unvaccinated would need to complete "three vaccination cycles (...), pushing them into September or October." [45].
28/01/2022	17/02/2022	28.01.2022: Half of the population "boostered" As per the Robert Koch Institute, over half the population in Germany, equating to at least 43 million individuals or 51.7% of the population, have now received a booster shot against COVID-19. Given the highly transmissible Omicron variant, this booster vaccination is crucial for providing effective protection. Up till now, a minimum of 75.6% of the population (approx. 62.9 million people) have received at least one dose of the vaccine. Full basic immunization, typically requiring two doses, has been administered to at least 73.7% of the population, translating to around 61.3 million individuals. [45]. 27.01.2022: Moderna: Clinical trials on omicron [45].
17/02/2022	18/03/2022	17.02.2022: Corona restrictions to fall until 20 March. The federal and state governments in Germany have decided on a phased rollback of COVID-19 measures over

		the forthcoming four weeks, culminating at the beginning of spring on March 20. The plan involves discontinuing all extensive protective measures, provided that the healthcare facilities can manage the situation. The only persisting "basic protection" will be the obligation to wear masks in indoor environments and on public transportation. [45].
18/03/2022	03/04/2022	18.03.2022 Bundestag passes abolition of nationwide Corona requirements. Despite elevated infection rates, the German Bundestag has approved revisions to the Infection Protection Act, leading to the termination of the majority of national COVID-19 protective measures. Beginning Sunday, only a small number of overarching requirements regarding masks and testing in facilities serving vulnerable groups will persist. Face covering will continue to be obligatory in buses and trains. Additional restrictions may be implemented for regional areas identified as "hotspots" if the state parliament declares a particularly critical COVID-19 situation in those areas. [45].
03/04/2022	26/04/2022	03.04.2022: Numerous Covid protective measures dropped. As of Sunday, the majority of regional COVID-19 restrictions impacting daily life were lifted across large swathes of Germany, even though infection rates are still significant. The new universal legal system imposes only a handful of overall protective measures. In nearly every federal state, masks are now required solely in healthcare facilities such as doctors' offices, nursing homes, and

		hospitals, as well as on buses and trains. Testing remains mandatory in schools, for instance. Masks are still obligatory on long-haul trains and airplanes across the country. Irrespective of governmental rules, businesses, stores, and other organizations can uphold their own guidelines, such as mandatory mask wearing, according to internal policies. The federal government has defended its decision to eliminate most of the protective measures, citing that the healthcare system is not at capacity and stricter regulations can still be implemented locally in the event of an emergency. [45].
26/04/2022	01/06/2022	Entry regulations to be extended until the end of May. To guard against the spread of the coronavirus, travel regulations for those returning from holidays or entering Germany are slated to continue for the foreseeable future. The Federal Ministry of Health plans to extend the entry regulation, which is about to expire, until the end of May, according to a ministry spokesperson. As a result, anyone aged 12 or above will still need to demonstrate upon entering Germany that they have been vaccinated, have recovered from Covid, or have tested negative. The rule also stipulates quarantine requirements for those returning from designated high-risk or virus-variant regions, although, at present, the Federal Government does not classify any country as such. [45].
01/06/2022	17/06/2022	Corona entry rules to be relaxed as of 1 June. Considering the decreasing COVID-19 case numbers, Federal Health Minister Karl Lauterbach is intending to ease the entry restrictions into Germany during the summer

		months. "We will suspend the 3G rule on entry until the end of August," the SPD politician told the newspapers of the Funke Mediengruppe. As of June 1, people traveling back to Germany or other immigrants will no longer need to provide evidence of vaccination, recovery from COVID-19, or a negative test result. The current Corona Entry Ordinance, which is in effect until May 31, mandates 3G proof for everyone over the age of twelve. [45].
17/06/2022	02/09/2022	17.06.2022: Seven-point plan for autumn Federal Health Minister Karl Lauterbach has urged caution in light of the recent uptick in COVID-19 cases, and is formulating a protection strategy for the upcoming fall season. A key element of this seven-point plan is a vaccination campaign that uses a variety of vaccines. The objective is to strategically close vaccination gaps, Lauterbach stated. Furthermore, a strategy is being developed to enhance the administration of medication to patients. [45].

02/09/2022		02.09.2022 After Ema recommendation: EU Commission approves Omikron vaccines The European Commission has approved two COVID-19 vaccines tailored to the Omicron variant, as announced by Health Commissioner Stella Kyriakides via Twitter on Friday. This follows a recommendation from the European Medicines Agency (EMA), with experts from the agency green-lighting applications from Biontech/Pfizer and the U.S. firm Moderna on Thursday. The Commission then took only a few hours to make its decision, according to Kyriakides. The vaccines are bivalent mRNA formulations that offer protection against both the original Sars-CoV-2 and the Omicron subline BA.1. Though BA.1 is currently insignificant in Germany, experts believe that these new vaccines could also have advantages against the presently dominant subtype BA.5 [45]..
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Table 5 Timeframe for cases

6.6 Building Case Library

Author: Asnel Christian (100%)

Upon determining the appropriate time frames, we proceed to construct our case library. This process is conducted programmatically, by aggregating the acquired data and isolating the columns of significance.

Given that our analysis incorporates time frames, it's paramount to ensure a comprehensive depiction of values. Consequently, we employ cumulative values and, where feasible, calculate average values to provide a more lucid understanding of the dataset under examination.

The subsequent methodology elucidates our process for computing averages, predicated on a given dataframe and a specified case time frame.

```
def get_avg_between_dates(df, date_col, start_date, end_date):|
    # convert the date column to datetime format
    df[date_col] = pd.to_datetime(df[date_col], format='%d-%m-%Y')

    # filter the rows based on the date range
    filtered_df = df[(df[date_col] >= start_date) & (df[date_col] <= end_date)]

    # calculate the average of all numeric columns
    numeric_cols = filtered_df.select_dtypes(include='number').columns.tolist()
    avg = filtered_df[numeric_cols].mean()
    return avg
```

Figure Script for averaging numerical values in dataframe given case's time frame.

The following code shows how we extract all the columns of interest given the computed data.

```
def build_final_df(town, date_col, start_date, end_date, icu_df, weather_df, death_df, final_df, vaccination_df):
    avg_weather = get_avg_between_dates(weather_df, 'Current Date', start_date, end_date)
    avg = get_avg_between_dates(icu_df, date_col, start_date, end_date)
    new_row = {
        'start_date': start_date,
        'end_date': end_date,
        'problem_start_number_of_icu_active_cases': icu_df[icu_df['date'] == start_date]['COVID_patients_in_ICU'].to_numpy()[0],
        'problem_end_number_of_icu_active_cases': icu_df[icu_df['date'] == end_date]['COVID_patients_in_ICU'].to_numpy()[0],
        'city': town,
        'problem_start_number_of_deaths': death_df[death_df['date'] == start_date]['deaths'].to_numpy()[0],
        'problem_end_number_of_deaths': death_df[death_df['date'] == end_date]['deaths'].to_numpy()[0],
        'problem_start_number_of_active_cases': death_df[death_df['date'] == start_date]['active_cases'].to_numpy()[0],
        'problem_end_number_of_active_cases': death_df[death_df['date'] == end_date]['active_cases'].to_numpy()[0],
        'problem_average_temperature': avg_weather['temperature'],
        'problem_vaccinated_population': vaccination_df[vaccination_df['Impfdatum'] == start_date]['population_percentage'].to_numpy()[0],
        'problem_average_humidity': avg_weather['relative_humidity'],
        'solution_lockdown_policy_level': stayhome[stayhome['Day'] == start_date]['stay_home_requirements'].to_numpy()[0],
        'solution_vaccine_policy_level': policies[policies['Date'] == start_date]['H7E_Vaccination policy'].to_numpy()[0],
        'solution_mask_policy_level': policies[policies['Date'] == start_date]['H6E_Facial Coverings'].to_numpy()[0],
    }
    final_df = final_df.append( new_row, ignore_index=True)
    return final_df
```

Figure Script for extracting columns of interest given computed data frames.

And finally, we implement for the following piece of code segment with the intent to extract information pertinent to each city initially, prior to constructing its associated case library.


```

def add_data_by_city(icu_name, weather_name, death_name, timeframes):
    icu = icu_all[icu_all['city'] == icu_name]
    icu['date'] = pd.to_datetime(icu['date'], format='%m/%d/%Y')
    start_date = pd_dates[0]
    end_date = pd_dates[-1]
    icu = icu[(icu['date'] >= start_date) & (icu['date'] <= end_date)]

    deaths_df = pd.read_csv(f'./{death_name}.csv')
    deaths_df['date'] = pd.to_datetime(deaths_df['date'], format='%Y-%m-%d')
    start_date = pd_dates[0]
    end_date = pd_dates[-1]
    deaths_df = deaths_df[deaths_df['date'].isin(pd_dates)]

    weather_df_koblenz = weather_df[weather_df['District Name'].str.endswith(weather_name)]
    weather_df_koblenz['Current Date'] = pd.to_datetime(weather_df_koblenz['Current Date'], format='%Y-%m-%d')
    weather_df_koblenz = weather_df_koblenz[(weather_df_koblenz['Current Date'] >= start_date) & (weather_df_koblenz['Current Date'] <= end_date)]

    df = {'start_date': [],
          'end_date': [],
          'problem_start_number_of_icu_active_cases': [],
          'problem_end_number_of_icu_active_cases': [],
          'problem_vaccinated_population': [],
          'city': [],
          'problem_start_number_of_deaths': [],
          'problem_end_number_of_deaths': [],
          'problem_start_number_of_active_cases': [],
          'problem_end_number_of_active_cases': [],
          'problem_average_temperature': [],
          'problem_average_humidity': [],
          'solution_lockdown_policy_level': [],
          'solution_vaccine_policy_level': [],
          'solution_mask_policy_level': []},
          }

    cases_df = pd.DataFrame(df)
    for i, _ in enumerate(timeframes[:-1]):
        cases_df = build_final_df(death_name, 'date', timeframes[i], timeframes[i + 1], icu, weather_df_koblenz, deaths_df, cases_df, vaccination)
    return cases_df

```

Figure Script for extracting data about a city and building corresponding case library.

Upon implementation of the aforementioned code segments, we are equipped to construct cases for cities within the Rheinland-Pfalz region, utilizing the data that has been gathered.

In this chapter, we discussed methodology to collect data, and collected data from different sources as well as making certain choices to collect missing data via a method we discussed earlier. We then discussed different approaches to build timelines for our CBR system to represent various cases during different triggering events, we ended the chapter by providing a list of the timelines. In the next chapter, we have a closer look at the architecture of our system in more detail.

7. System Architecture

Authors: Shehab (30%), Shuvam (70%)

On this section we focus on the architecture of the case-based reasoning expert system integrated with rules, to make the solution more personalized to the immediate case i.e., it uses rules and facts as an additional adaptation mechanism.

The expert system's main goal is to facilitate policy-level decision-making in the context of dealing with pandemic like crisis, in our context COVID-19 outbreak. The system takes advantage of prior experiences to inform policy recommendations for situations comparable to the past by leveraging a large case base of historical incidents and their associated solutions.

The architecture of the system presented here consists of various components, including the case base reasoner module, case adaptation module (rule engine), explanatory module, case library, case acquisition module and the user interfaces. The whole components work in harmony to facilitate the retrieval of similar cases from the case library, the execution of rules for solution adaptation, using explanatory system to explain the solution in more human understandable manner and then presentation of recommendations to decision-makers.

Below we present the architecture of the system and explain its flow and all relationships between the components.

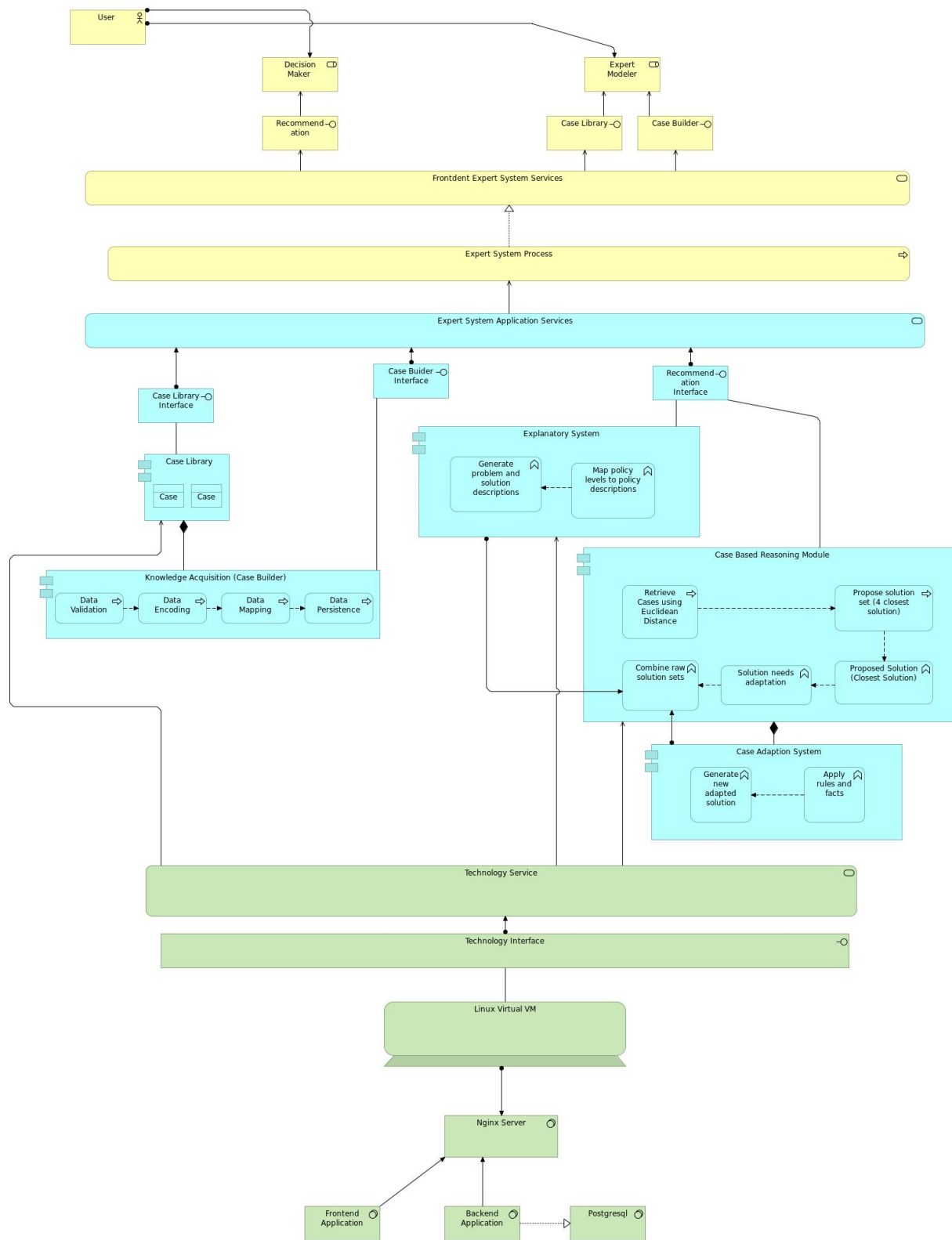
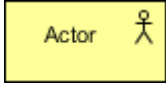
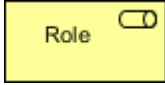
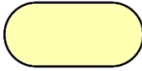
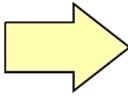
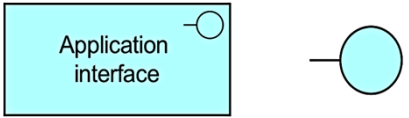
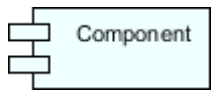
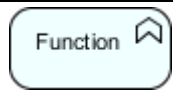
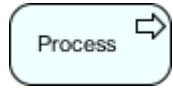
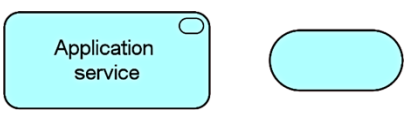
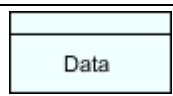


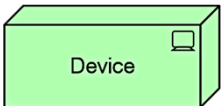
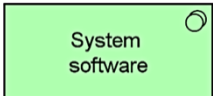
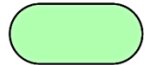
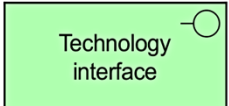
Figure 11 System Architecture

Component Legends

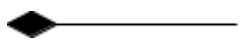
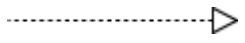
The component and relationship legends presented below are according to [46].

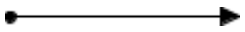
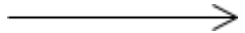
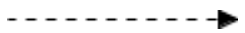
Name	Representation	Definition
Actor		An entity that is capable of performing behavior.
Role		The responsibility for performing specific behavior, to which an actor can be assigned, or the part an actor plays in a particular action or event.
Business Interface	 	A business interface represents a point of access where a business service is made available to the environment.
Business Services	 	A business service represents explicitly defined behavior that a business role, business actor, or business collaboration exposes to its environment.
Business Process	 	A business process represents a sequence of business behaviors that achieves a specific result such as a defined set of products or business services.

Application Interface		A point of access where application services are made available to a user, another application component, or a node.
Application Component		An encapsulation of application functionality aligned to implementation structure, which is modular and replaceable. It encapsulates its behavior and data, exposes services, and makes them available through interfaces.
Application Function		Automated behavior that can be performed by an application component.
Application Process		A sequence of application behaviors that achieves a specific outcome.
Application Service		An application service represents an explicitly defined exposed application behavior.
Application Data		Data structured for automated processing.
Technology Device		A device represents a physical IT resource upon which system

	 	software and artifacts may be stored or deployed for execution.
Technology System Software	 	System software represents software that provides or contributes to an environment for storing, executing, and using software or data deployed within it.
Technology Service	 	A technology service represents an explicitly defined exposed technology behavior.
Technology Interface	 	A technology interface represents a point of access where technology services offered by a node can be accessed.

Relationship Legends

Name	Representation	Definition
Composition		Indicates that an element consists of one or more other concepts.
Realization		Indicates that an entity plays a critical role in the creation,

		achievement, sustenance, or operation of a more abstract entity.
Assignment		Expresses the allocation of responsibility, performance of behavior, or execution.
Serving		Models that an element provides its functionality to another element.
Flow		Transfer from one element to another.
Association		Models an unspecified relationship, or one that is not represented by another ArchiMate relationship.
Junction	  (And) Junction (Or) Junction	Used to connect relationships of the same type.

We divide this architecture into three main architectural layers business, application, and technology layers. A user interacts with these layers.

7.1 Users

Users are the people who will use the recommendation system which has roles. We divide them into two main users, a decision maker, and an expert modeler. Each has different responsibilities to perform. Although the prototype does not explicitly distinguish between them internally but here, we show them as two users. This is done to avoid complexity with authorizations internally, but it could be considered as a feature in the future.

Decision Maker

A decision maker is whom the system is intended to use. Usually, these are policy makers who are looking for recommendations regarding the pandemic. A decision maker will typically use the recommendation interface to fill in some required information and will receive a descriptive and textual recommendation informing them of the policies to implement.

Expert modeler

An expert modeler is typically a person responsible for maintaining the case library by adding new cases to cover more situations, which in turn increases the knowledge base with up-to-date information. As well as accessing the case library.

7.2 Business Layer

Recommendation interface:

This interface is what the user sees when looking to get a recommendation. The interface is presented as a form where the user can enter information such as Start Date (required), End Date, City (required), Number of Active Cases at Start (required), Number of Active Case at End, Number of ICU Active Cases at Start (required), Number of ICU Active Cases at End, Number of Death at Start (required), Number of Death at End, Vaccinated Population, Average Temperature, Average Humidity.

Case Builder interface:

This interface is what the system admin sees when updating the case library. The interface is presented as a form where the user can enter information such as Start Date End Date City Number of Active Cases at Start Number of Active Case at End, Number of ICU Active Cases at Start, Number of ICU Active Cases at End, Number of Death at Start, Number of Death at End, Vaccinated Population, Average Temperature, Average Humidity, Lockdown Policy, Mask Policy, Vaccination Policy. All these form fields are required to build a new case.

Case Library interface:

This interface is what the users can use to view cases in the case library. The interface is presented as a table with columns and rows corresponding to the properties of the cases.

7.3 Application Layer

Recommendation interface:

This interface represents an endpoint in the API which the client talks to when sending a request from the same interface in the business layer to get a recommendation.

Case Builder interface:

This interface represents an endpoint in the API which the client talks to when sending a request from the same interface in the business layer to add a new case to the case library.

Case Library interface:

This interface represents an endpoint in the API which the client talks to when sending a request from the same interface in the business layer to view all the cases in the case library.

Knowledge Acquisition (Case Builder):

The Case Builder is a module in the system, which is responsible for validating, encoding, mapping, and persisting the data coming to the case library. Validation is the process of making sure the input is in the correct format type and the required data has been sent. Encoding is the process of converting categorical values from text to number for easier distance calculation as well as calculating things such as infections rate, and death rate. Mapping is the process of mapping the right fields to the right properties in the database as well as assigning the right data to a given city. This means that if Koblenz is the city, the system maps its population and population density to the city. Persistence is the processing of storing the incoming data after it has gone through the previous three steps.

Explanatory System:

Our expert system's Explanatory System is essential to improving the readability and understanding of the system's suggested solutions. Its main purpose is to convert the numerical values contained in the case structure, including the solution, into concise phrases that can be understood by humans by following a defined sentence pattern. To maintain consistency and intelligibility across all generated descriptions, the Explanatory Module follows a standardized description template. It uses simple words, follows grammatical rules and, and tries to communicate the important information clearly.

Additionally, it also performs similar tasks to add problem description in each new case. There are three main components essential in this module:

1. Mapping policy level to policy description:

The Explanatory Module transforms numerical value from the case structure policy solutions into insightful descriptions using predefined mappings and description template. It creates a mapping relationship between policies of the case structure, with the predefined standardized description with respect to their associated terms, resulting in the standardization and consistency of sentence generation.

2. Problem description:

The Explanatory Module is responsible for transforming the numeric values representing the case structure into a comprehensible statement. It maps the relevant case characteristics, such as the number of active cases, severe cases, deaths, vaccination status, and inflation rate, to descriptive phrases that provide a concise summary of the problem context in problem description.

3. Case description:

Similarly, the Explanatory Module creates a succinct and clear sentence summarizing the suggested course of action when a solution is recommended from the system in solution description. The result is a concise and well-organized policy statement that blends pertinent numerical numbers from the solution with the predetermined sentence structure.

Case Based Reasoning Module:

The system's core component, the Case-Based Reasoning (CBR) module, uses Euclidian distance metric to generate solutions based on previous case data. To find the cases that are the most similar to the current case in hand, it compares the user's case with the cases kept in the case library. Taking a closer look at the CBR module's main features:

1. Retrieve Cases using Euclidian Distance:

After the data is entered through user interface and forwarded to the CBR module, the CBR module uses Euclidean distance metric, to determine how similar the new case is with respect to cases in the case library. Number of active cases, the number of severe cases, the number of deaths, the inflation rate, location, the mask policy, and the vaccine policy are among the some of the relevant

case features that are considered for determining the distance between the cases. The distances are then aggregated to obtain an overall similarity measure for each case in the library.

2. Proposed Solution Set:

The case library is a collection of cases that have already been dealt with; each case includes comprehensive information about the past COVID-19 situations, policies, and solutions. The CBR module retrieves the top 4 cases from the case library that closely match the user's input case. By retrieving multiple similar cases, the CBR module provides a broader perspective of solutions, capturing a range of similar scenarios and their solutions. It considers various case features using distance metric to make the retrieval. The retrieved cases represent past scenarios that are most similar to the current situation.

3. Top Recommendation:

Among the top four similar cases retrieved only the most similar cases becomes the top recommendation, suggesting the solution that aligns most closely with the user's specified scenario.

4. Solution Needs Adaptation (Validation of Solution):

The CBR module performs a recommendation validation step to determine whether the priority recommendation from the system is acceptable as the final recommendation. Currently, the prototype system does not have complete data regarding all the possible scenarios regarding all the policies implementations; the prototype system fails to recommend a valid solution, hence why case adaptation was introduced. This verifies if the recommended solution meets some of the predefined conditions of desirability. If the solution meets the validation conditions, it is considered as the final solution and forwarded to the explanatory module and if the solution doesn't meet the validation conditions or is not deemed the best recommendation, it is forwarded to the case adaptation module for further analysis and refinement.

Case Adaptation System:

In certain scenarios where the top recommendation from the system does not meet the validation conditions or is not considered the best recommendation for that specific case, the CBR module forwards the top recommendation to the Case Adaptation System. The Case Adaptation System further analyzes the recommendation and applies the rule-based technique to adapt, refine and

make the solution more personalized to that specific scenario. After modifying the recommendation, it is then forwarded to the Explanatory System for further processing.

Rules for Case Adaptation:

These rules are intended to make the CBR system's proposed solution more adaptable and personalized. The aim is to create accurate results. In other words, these rules serve as an optimization mechanism.

1. Lockdown Policy Evaluation rule:

- a. This rule evaluates the lockdown policy level of the case based on the provided parameters: level (current lockdown policy level), start_cases (number of active cases at the start), and population (total population of that place during that time period).
- b. The rule then calculates the cases_percentage (i.e., total number of the active cases divided by total population of that place then multiplied by hundred) relative to the population.
- c. If the cases_percentage falls within the range of 0.32 to 0.45 and the current lockdown policy level (level) is either 2, 1, or 0, the adaptation module suggests a lockdown policy level of 2.
- d. And if the cases_percentage exceeds 0.45 and the current lockdown policy level (level) is either 2, 1, or 0, the adaptation module suggests a lockdown policy level of 3.
- e. Otherwise, if none of the above conditions are met, the adaptation module suggests a lockdown policy level of 0, indicating no lockdown measures are to be implemented.

2. Vaccine Policy Evaluation rule:

- a. This rule determines the vaccine policy description based on the given conditions.
- b. If the current case in hand problem vaccinated population is 0 (indicating no vaccine has been issue), it assigns the vaccine policy description using a function that categorizes the vaccine policy.
- c. And, if the solution vaccine policy level of the recommended solution (similar_data["solution_vaccine_policy_level"]) is less than the current case in hand problem vaccinated population value, it assigns the vaccine policy description using the same function as above, considering the value of current case in hand problem vaccinated population.

- d. Otherwise, if none of the above conditions are met, it assigns the vaccine policy description using the same function, considering the value of recommended solution (`similar_data["solution_vaccine_policy_level"]`).

7.4 Technology Layer

Linux Virtual VM:

This virtual machine hosts the recommender system on the university's cloud environment. It includes all the runnable code for the frontend and backend of the application as well as the database.

Frontend Application:

This is the user interface of the application which is what the users will interact with when visiting the system online.

Backend Application:

The backend is the API portion of the application which exposes endpoints for the frontend. It includes the major logical components discussed earlier such as case-based reasoning module, explanatory system, knowledge acquisition, and case adaptation system. Most of which query the database for information.

PostgreSQL:

This is the database we used to store the data. It stores data related to the recommendations produced by the system, and cases in separate tables. It is conceptually represented in the modeled diagram as the Case Library.

In this chapter, we showed the architecture of the system to show cases the flow of the system, components, application, technologies, and users in the system. The chapter discussed in detail each layer of architecture. In the next chapter, we discuss the implementation of the architecture discussed here.

8. System Implementation

Authors: Shehab (40%), Sudan (60%)

This chapter showcases the case-based reasoning system built during this project. It explains the system flow, the user interface, and the API, as well as showing the development process behind it and the technology used.

8.1 Tech Stack

Tech stack refers to the sets of technologies used to build applications. In this case the application is the recommendation system. Here we explore the technology behind the system implementation. For the user interface also known as frontend, we used a user interface library called React with Material UI as a component library to compose readymade user interfaces (UI)s layouts as well as JavaScript as the main frontend language. In the backend we used Python primarily in combination with FastAPI which is a backend framework along with PostgreSQL as the choice of database for persisting cases and recommendations. Lastly, Nginx is used as the web server to serve the application from its deployment machine.

8.2 System Flow and User Interface

The case-based reasoning system consists of four main UI layouts, recommendation, case builder, case library as well as an about page. In the architecture chapter we explained how the architecture works and what kind of users, interfaces and modules exist. Here we will show how the UI of the system interacts with the backend modules to showcase the flow of the system. It is important to note that the system validates the input entered in the frontend and the backend, however, it is limited to only validating the type of data entered not the truthfulness of the data as this is outside of the scope for this prototype. This means that for example, if the system requires in a certain field a number to be entered but the user entered a text, the system will inform the user that the type entered is wrong. However, the system will not validate if temperature matches humidity entered for the reasons explained earlier.

Recommendation Page:

The recommendation is the main interface the user sees when they visit the system. The page includes two things, an explanation of what this page does and a form which includes fields for

the user asking for recommendation to fill. These form fields include, Start Date (required), End Date, City (required), Number of Active Cases at Start (required), Number of Active Case at End, Number of ICU Active Cases at Start (required), Number of ICU Active Cases at End, Number of Death at Start (required), Number of Death at End, Vaccinated Population, Average Temperature, Average Humidity. After they enter this information, they can submit this request by clicking on the submit button at the end of the form. The backend takes this request and runs it through the CBR module which processes this request by doing a similarity check between this new case and previous cases. Then the system recommends four solutions, the closest solution is then picked for further adaptation if needed. The system then takes the closest solution as well as the three other closest solutions. All the solutions as well as the main recommendation are presented in a textual manner. Refer to the figures below to see how this page looks before and after the recommendation.

Figure 12 Recommendation page before recommendation

Recommendation

In a scenario where the population density is **moderate**, median age is **44.5**, infection rate is **0.18** percentage, and mortality rate is **0.09** percentage, with **12** ICU active cases. It is recommended to implement a **level 2 lockdown policy**, (require not leaving house with exceptions for daily exercise, grocery shopping, and essential trips).

A **level 2 mask policy** (required mask in some specified shared/public spaces outside the home with other people present, or some situations when social distancing not possible).

It is important to note that (vaccination available for all the following: key workers/ clinically vulnerable groups / elderly groups).

Overall, taking these measures can help control the spread of the disease and minimize the number of severe cases and deaths in the affected area.

Similar cases:

Case 1: In a scenario where the population density is **low**, median age is **45.7**, infection rate is **0.16** percentage, and mortality rate is **0.11** percentage, with **2** ICU active cases. It is recommended to implement a **level 0 lockdown policy**, (require not leaving house with exceptions for daily exercise, grocery shopping, and essential trips).

A **level 2 mask policy** (required mask in some specified shared/public spaces outside the home with other people present, or some situations when social distancing not possible).

It is important to note that (vaccination available for all three, plus partial additional availability select broad groups/ages).

Overall, taking these measures can help control the spread of the disease and minimize the number of severe cases and deaths in the affected area. In a scenario where the population density is **low**, median age is **44.7**, infection rate is **0.17** percentage, and mortality rate is **0.1** percentage, with **0** ICU active cases. It is recommended to implement a **level 0 lockdown policy**, (no lockdown measures).

A **level 2 mask policy** (required mask in some specified shared/public spaces outside the home with other people present, or some situations when social distancing not possible).

It is important to note that (vaccination available for all three, plus partial additional availability select broad groups/ages).

Overall, taking these measures can help control the spread of the disease and minimize the number of severe cases and deaths in the affected area. In a scenario where the population density is **low**, median age is **45.7**, infection rate is **0.15** percentage, and mortality rate is **0.05** percentage, with **5** ICU active cases. It is recommended to implement a **level 2 lockdown policy**, (require not leaving house with minimal exceptions (e.g., allowed to leave only once every few days, or only one person can leave at a time, etc.) not implementing any lockdown policy).

A **level 2 mask policy** (required mask in some specified shared/public spaces outside the home with other people present, or some situations when social distancing not possible).

It is important to note that (vaccination available for all the following: key workers/ clinically vulnerable groups / elderly groups).

Overall, taking these measures can help control the spread of the disease and minimize the number of severe cases and deaths in the affected area.

Figure 13 Recommendation result

Case Builder Page:

The case builder page plays a vital role in the system as it is the place where a system admin can visit and add new cases to keep the case library up to date with new cases. It essentially serves as the knowledge collector page. A case added through this page is stored in the database. The page contains information about the case builder as well as form which include the following field all of which are required, Start Date End Date City Number of Active Cases at Start Number of Active Case at End, Number of ICU Active Cases at Start, Number of ICU Active Cases at End, Number of Death at Start, Number of Death at End, Vaccinated Population, Average Temperature, Average Humidity, Lockdown Policy, Mask Policy, Vaccination Policy. When this information is sent the system it gets handled by the knowledge acquisition module, which does preprocessing steps before saving the case and add new information to it such as infection rate, death rate, population, and population density. Refer to the figure below to see the page.

Case Builder

The case builder allows you to create a case which will be stored in the case library. The case library is the accumulated knowledge which the system uses to make recommendations. Use the form to build a case and then click the "Submit" button to add the case to the library.

Start Date
mm/dd/yyyy

End Date
mm/dd/yyyy

City

Number of Active Cases at Start

Number of Active Cases at End

Number of ICU Active Cases at Start

Number of ICU Active Cases at End

Number of Deaths at Start

Number of Deaths at End

Vaccinated Population

Average Temperature (Celsius) During Sel...

Average Humidity (%) During Selected Per...

Lockdown Policy

Mask Policy

Vaccination Policy

SUBMIT

Figure 14 Case Builder Page

Case Library Page:

The case builder page is intended to display records of the historical cases collected which are also used for the similarity distance measure. The page displays these records in a tabular format for easier navigation and distinction between the columns. It also paginates these records because the case library includes 860 historical cases collected from various cities in Rhineland-Palatinate. Refer to the figure below to see this page.

Case Library
The case library is the accumulated knowledge which the system uses to make recommendations. The table below shows the cases in the library. The cases are sorted by the date they were added to the library.

Case ID	Start Date	End Date	City	Population	Population Density	Median Age	Problem Description	Number of Active Cases at Start
1	2020-04-25	2020-05-15	Birkenfeld	80849	Low 300 or less	47.2	VIEW	27
2	2020-05-15	2020-06-01	Birkenfeld	80849	Low 300 or less	47.2	VIEW	9
3	2020-06-01	2020-06-30	Birkenfeld	80849	Low 300 or less	47.2	VIEW	5
4	2020-06-30	2020-07-01	Birkenfeld	80849	Low 300 or less	47.2	VIEW	3
5	2020-07-01	2020-07-30	Birkenfeld	80849	Low 300 or less	47.2	VIEW	3
6	2020-07-30	2020-08-01	Birkenfeld	80849	Low 300 or less	47.2	VIEW	5
7	2020-08-01	2020-08-31	Birkenfeld	80849	Low 300 or less	47.2	VIEW	5
8	2020-08-31	2020-09-01	Birkenfeld	80849	Low 300 or less	47.2	VIEW	18
9	2020-09-01	2020-09-30	Birkenfeld	80849	Low 300 or less	47.2	VIEW	18
10	2020-09-30	2020-10-01	Birkenfeld	80849	Low 300 or less	47.2	VIEW	17

< 1 2 3 4 5 ... 86 >

Figure 15 Case Library Page

8.3 API

This part focuses on the creation of an API using Python's FASTAPI framework, which is intended to create frictionless communication between a project's frontend and backend components. This API promotes effective and safe data sharing, enabling real-time interactions and improving user experience by utilizing the potent features of FASTAPI, a high-performance web framework. Additionally, a PostgreSQL database, a stable and scalable relational database management system, is seamlessly connected with the API to guarantee the storing of structured data in a trustworthy manner. It does this by combining FASTAPI with PostgreSQL. The major three APIs used in this application are as follows:

- **All Case:** The `/cases` API makes it easy to retrieve comprehensive historical case records stored in the database 'cases' table. This API uses the HTTP GET method to retrieve all the cases stored in the 'cases' table and returns the response in the form of a JSON data structure.
- **Create cases:** The `/cases/create` API is used to generate new historical cases. The payload is sent to the backend via the HTTP POST method. The new cases are added to the database's 'cases' table by the backend when it receives the payload in the form of a JSON file. By

implementing input validation into the */cases/create* API, the system makes sure that only legitimate and proper data type is allowed, increasing data consistency, reliability, and correctness within the database's 'cases' table.

- **Recommendation:** The */recommendation* API serves the purpose of providing recommendations based on user input. It employs the HTTP POST method to send the payload containing the user's input to the backend. The user's input is accepted in JSON format, allowing for a flexible and structured representation of the input data.

Upon receiving the payload, the API utilizes the input to calculate similarity measures, which are employed to determine the most relevant recommendations. The similarity measure used in this API is Euclidean distance. Once the calculations are performed, the generated recommendations are saved in the 'recommendation' table within the database. This ensures that the recommendations are persistently stored and can be accessed for future reference.

Furthermore, the API returns a recommendation and top three most similar historical cases solutions as reference to the recommendations to the user. This response provides valuable insights and suggestions based on historical cases that exhibit similarity to the user's input. By offering these references, the */recommendation* API assists users in making informed decisions and leveraging past experiences to address their current needs or concerns.

8.4 Deployment

The project prototype was made available on a web server with the IP address 141.26.157.102.

Prerequisites

Indeed, deploying the application typically requires certain prerequisites to ensure a successful deployment. Here are some of the prerequisites to consider when deploying the FAST API:

- A server with Ubuntu
- Python 3.10 or newer
- *sudo* access to your server via SSH
- PostgreSQL as a database
- Node version 18 or newer

Setup the App

To set up the application, begin by creating a directory for the project.

```
mkdir /home/cbr-rest-api-fastapi
```

Then navigate to the newly created directory.

```
cd /home/cbr-rest-api-fastapi.
```

Next, install the Python virtual environment by updating the package lists with

```
sudo apt update
```

Installing python3-venv.

```
sudo apt install python3-venv.
```

Afterwards, initialize and activate the virtual environment by running.

```
python3 -m venv venv
```

Activation of virtual environment.

```
source venv/bin/activate.
```

Now, you can add the project from GitHub.

<https://gitlab.uni-koblenz.de/rlab-ws-covid-and-ai-g1/case-based-reasoning-recommender.git>

Once added, install the required packages by executing

```
pip install fastapi uvicorn gunicorn
```

```
pip install -r requirements.txt
```

Creating database and configuration of .env file

Create a database on PostgreSQL with a username and password.

Rename the file `.env.example` to `.env` from the project directory and configure the env file according to the credentials of your database.

```
1 DATABASE_PORT=6500
2 POSTGRES_PASSWORD=your_db_password
3 POSTGRES_USER=your_db_username
4 POSTGRES_DB=your_db_name
5 POSTGRES_HOST=postgres
6 POSTGRES_HOSTNAME=127.0.0.1
```

Figure 16 .env file example

Creating System service for running the application

Making an application a system service makes it simple to start, restart, and stop the program. Another benefit is that if the server reboots, the system will launch the application automatically. We will make use of the gunicorn package that was set up in the preceding section to handle the service.

To create the *gunicorn_conf.py* file in the application root directory */home/cbr-rest-api-fastapi*, you can use the following command:

```
sudo nano /home/cbr-rest-api-fastapi/gunicorn_conf.py
```

Now the code below should be used in the *gunicorn_conf.py*

```
1  # gunicorn_conf.py
2  from multiprocessing import cpu_count
3
4  bind = "127.0.0.1:8000"
5
6  # Worker Options
7  workers = cpu_count() + 1
8  worker_class = 'uvicorn.workers.UvicornWorker'
9
10 # Logging Options
11 loglevel = 'debug'
12 accesslog = '/home/cbr-rest-api-fastapi/access_log'
13 errorlog = '/home/cbr-rest-api-fastapi/error_log'
14
```

Figure 17 Gunicorn configuration file

After creating the *gunicorn_conf.py* file, you'll need to save and exit the text editor. Once that's done, you can proceed to create a systemd unit file. Execute the following command to create the file:

```
sudo nano /etc/systemd/system/fastapi_cbr.service
```

Then copy the code below to the *fastapi_cbr.service* file:

```

[Unit]

Description=Gunicorn Daemon for FastAPI CBR

After=network.target


[Service]

User=ubuntu

Group=www-data

WorkingDirectory=/home/cbr-rest-api-fastapi

ExecStart=/home/cbr-rest-api-fastapi/venv/bin/gunicorn -c gunicorn_conf.py main:app


[Install]

WantedBy=multi-user.target

```

Figure 18 service file for the application

After creating the service file, you will need to start and enable the service using the following commands:

```
sudo systemctl start fastapi_cbr
```

```
sudo systemctl enable fastapi_cbr
```

To check if the service is enabled and running use the following command.

```
sudo systemctl status fastapi_cbr
```

If there is no problem the output will be:

```

ubuntu@AI-COV-RL5:/etc/systemd/system$ sudo systemctl status fastapi_cbr
● fastapi_cbr.service - Gunicorn Daemon for FastAPI CBR
   Loaded: loaded (/etc/systemd/system/fastapi_cbr.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2023-05-16 21:02:20 UTC; 2 days ago
     Main PID: 704 (gunicorn)
        Tasks: 41 (limit: 9491)
      Memory: 649.5M
         CPU: 20min 41.062s
       CGroup: /system.slice/fastapi_cbr.service
              └─704 /home/cbr-rest-api-fastapi/venv/bin/python3 /home/cbr-rest-api-fastapi/venv/bin/gunicorn -c gunicorn_conf.py main:app
                └─802 /home/cbr-rest-api-fastapi/venv/bin/python3 /home/cbr-rest-api-fastapi/venv/bin/gunicorn -c gunicorn_conf.py main:app
                  └─803 /home/cbr-rest-api-fastapi/venv/bin/python3 /home/cbr-rest-api-fastapi/venv/bin/gunicorn -c gunicorn_conf.py main:app
                    └─804 /home/cbr-rest-api-fastapi/venv/bin/python3 /home/cbr-rest-api-fastapi/venv/bin/gunicorn -c gunicorn_conf.py main:app
                      └─805 /home/cbr-rest-api-fastapi/venv/bin/python3 /home/cbr-rest-api-fastapi/venv/bin/gunicorn -c gunicorn_conf.py main:app
                        └─806 /home/cbr-rest-api-fastapi/venv/bin/python3 /home/cbr-rest-api-fastapi/venv/bin/gunicorn -c gunicorn_conf.py main:app

May 16 21:02:20 AI-COV-RL5 systemd[1]: Started Gunicorn Daemon for FastAPI CBR.
ubuntu@AI-COV-RL5:/etc/systemd/system$

```

Figure 19 Service status output

Installing and Configuring Nginx

Nginx acts as a reverse proxy server, forwarding client requests to the appropriate backend application (in our case, FastAPI application) and serving the application's responses back to the clients. Nginx is commonly used for load balancing, caching, and serving static files as well.

Additionally, Nginx can be configured to handle domain names and set up virtual hosts. This allows us to associate our FastAPI application with a specific domain name, enabling users to access the application using that domain.

Installing the nginx package.

```
sudo apt install nginx
```

Then we need to start and enable the nginx.

```
sudo systemctl start nginx
```

```
sudo systemctl enable nginx
```

After Nginx is installed and enabled, we need to configure it to serve our application. For this following command and code is required:

```
sudo nano /etc/nginx/sites-available/api.cbr.dev
```

Add the following code to the file *api.cbr.dev*.

```

server {
    client_max_body_size 64M;
    listen 80;
    server_name http://cbr-api.dev;

    location / {
        proxy_pass          http://127.0.0.1:8000;
        proxy_read_timeout  60;
        proxy_connect_timeout 60;
        proxy_redirect       off;

        # Allow the use of websockets
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
    }
}

```

Figure 20 Nginx configuration file

After saving the Nginx configuration, you'll need to enable the configuration and restart the Nginx service to apply the changes. To do this, you can use the following command:

```
Sudo ln -s /etc/nginx/sites-available/api.cbr.dev /etc/nginx/sites-enabled/
```

```
sudo systemctl restart nginx
```

Now the backend application is hosted and running.

Frontend Deployment

Navigate to client directory following command can be used:

```
cd /home/ cbr-rest-api-fastapi/client
```

Install the npm packages.

```
npm install
```

Build the project.

```
npm run build
```

Now, the application is ready and running and can be accessed through <http://141.26.157.102/> under the university of Koblenz network.

In this chapter, we looked deeply at the technical aspect of the system. This included the technology stack used to build out the system. We also showed how the system UI looks and how it works as well as discussed, the important API endpoints in the system. We ended the chapter by discussing the deployment process of the system for future reference.

9. Alternative Approach

This chapter discusses the different approaches the team took to implement the recommendations system. Based on our initial research, we looked at several approaches including the use of expert systems and their types, and machine learning. Here, we highlight three main approaches that were taken initially before considering CBR primarily. The code generated from these approaches can be found in the cloud in a folder named “Alternative Approaches”.

9.1 Rule Based Approach

Author: Shuvam (100%)

A rule-based decision-making expert system is a type of artificial intelligence (AI) that uses a knowledge base of rules and logical reasoning to make decisions. These systems are designed to mimic the decision-making process of a human expert in a particular field, such as medicine or policies.

Similar to other types of expert systems, Rule Based Expert System mainly consists of three main components: Knowledge Base, Inference Engine, and User Interface. In addition, there is a Knowledge Acquisition System and Explanatory Module.

Findings and recommendation:

The expert system that we proposed as alternatives were a standalone system for either CBR or Rule-base, or a hybrid mix of both. While working on a rule-based expert system as an alternative we developed a working prototype for the rule-based system. It takes 5 factors/features into account .i.e., age distribution, population density, infection rate, number of severe new cases, and number of deaths, before making any decision regarding mask, stay-home restriction and vaccination policy.

All the possible case scenarios can be taken into account if those 5 factors are the only important information that is required to recommend a policy (.i.e., somewhere around more than 600+ rules). If new added information is required to make the recommendation more accurate the rules are needed to be updated and further research are needed to be done to determine how the new factors that are being added affect the policy after being implemented to make our reasoning valid and accurate for new cases.

Developing a rule-based expert system for the COVID-19 pandemic would require extensive research to identify the most effective policies and recommendations. This would involve analyzing the correlation between different policies that were implemented in the past and their results in terms of reducing the number of cases, serious cases, deaths, and infection rates. This would help to reason/justify why some rules or policies were recommended during certain scenarios. Without this research, the system may not be able to generate accurate and effective policies and recommendations. This is the only way to justify why some rules were implemented on certain scenarios.

While a working prototype of a rule-based expert system for the COVID-19 pandemic has been developed, further research is required to make the system recommendation more perfect. This would require analyzing the data and refining the rules used in the system. However, due to the time constraint and collective decision from the group and the supervisors we have decided to move forward with standalone CBR system with minimal usage of rule for case adaptation. We suggest the establishment of a new research lab focused on the findings of this research lab for a rule-based system. Our current research lab has developed a functional prototype, but further research is required to refine the system and make it more the recommendation more efficient and personalized. A dedicated research lab would be invaluable in analyzing past cases and identifying which policies were most effective in reducing the number of cases, serious cases, deaths, and infection rates so that it could be integrated a possible recommendation for similar scenarios in future. Such a lab would make a significant contribution to the field and potentially save many lives. Python implementation of this system is uploaded with filename, 'alternate_approach_rule_based_system.ipynb'.

Example Rules:

[Disclaimer: All the recommendation that are given for the example are hypothetical and in-depth research is required for more accurate and justifiable recommendation.]

Rule 1.1:

IF (population density is high) AND (age distribution is Older Population) AND (infection rate is high) AND (number of deaths is high) AND (number of severe cases is high) THEN (lockdown policy should be at least level 3) AND (mask policy should be at level 4) AND (vaccine policy should be at level 3)

Explanation: In a scenario where the population density is high, age distribution is older population, infection rate is high, death is high, and there are high severe active cases, a stricter lockdown policy of not leaving house with minimal exceptions is needed to contain the spread. Additionally, a higher-level mask policy that requires personals to use masks outside the home at all times, regardless of location or presence of other people, and vaccine policy of providing vaccine to at least key-workers, clinically vulnerable groups and elderly should be implemented to reduce transmission and protect vulnerable populations.

Rule 1.2:

IF (population density is high) AND (age distribution is younger population) AND (infection rate is high) AND (number of deaths is low) AND (number of severe cases is low) THEN (lockdown policy should be at least level 1) AND (mask policy should be at level 2) AND (vaccine policy should be at level 4)

Explanation: In a scenario where the population density is high, age distribution is younger population, infection rate is high, death is low, and there are low numbers of severe active cases, a loose lockdown policy with less restrictions is recommended. Additionally, a moderate mask policy where people need to use mask in some specified spaces or situation should be implemented, and vaccine for all population policy should also be implemented to reduce transmission and protect vulnerable populations.

9.2 Machine Learning Approaches

Here we discuss the usage of standalone machine learning algorithms as a recommendation mechanism as well ensemble learning.

Ensemble Learning:

Author: Shehab (100%)

Ensemble learning is an approach in machine learning where multiple algorithms are combined to achieve a better predictive result via a voting schema [47]. In our case, we combined three different algorithms which were discussed in the background research chapter, Decision Trees, SVM and KNN. The case library that we produced earlier was used as training data to predict a policy for a new problem. In our testing, the ensemble learning approach had good results, but it was equivalent to Decision Trees alone recommendations. However, due to the limited time and research scope,

we decided not to continue with this approach and could be used or continued by a different research lab.

Decision Trees:

***Author:** Sudan (100%)*

Decision trees use a tree like model to make classification and regression and its used divide and conquer. In big databases, decision trees can be used to find characteristics and extract patterns that are crucial for discriminating and predictive modeling [48]. In our case, we applied the decision tree algorithm in our cases library to predict the policy levels. However, the result from the decision tree were like our initial approach of using similarity measure.

KNN:

***Author:** Sudan (100%)*

K-Nearest Neighbors (KNN) is a classification algorithm in machine learning that assigns labels to new data points by comparing their similarity to existing data points in the training set, using the majority class of the k nearest neighbors as the basis for classification [49]. Trying out a different approach for generating recommendation based on our case library. The KNN algorithm did not perform well. The predicted policies from the KNN algorithm were not reliable.

Furthermore, to determine which method should be used in for the CBR we tested three methods similarity measure (Euclidian distance), Decision tree and KNN by feeding the algorithm with the same data and analyzing the outcome. Where similarity measure and Decision tree have the same output. KNN had a different output to the same data and the recommendation of policies were not right as it was returning Level 0 lockdown policy in every test case. To support the testing and choosing the best method we did an evaluation metrices for KNN and Decision tree.

KNN	solution_vaccine_policy_level					Decision Tree	solution_vaccine_policy_level				
		precision	recall	f1-score	support			precision	recall	f1-score	support
	0	0.60	0.85	0.70	40	0	1.00	1.00	1.00	40	
	2	0.12	0.17	0.14	12	2	0.80	1.00	0.89	12	
	3	0.30	0.12	0.18	24	3	1.00	0.88	0.93	24	
	5	0.78	0.67	0.72	54	5	1.00	1.00	1.00	54	
	accuracy			0.58	130		accuracy			0.98	130
	macro avg			0.45	130		macro avg			0.95	130
	weighted avg			0.57	130		weighted avg			0.98	130

Figure 21 Evaluation metrics for KNN and decision tree

In the above figure we can see the KNN model accuracy is 0.58 which is lower than the accuracy of decision tree modal which is 0.98. Decision tree outperformed the KNN modal in every evaluation metrics i.e., precision, recall and f1-score. However, due to the same result from decision tree and our initial approach we decided to move forward with the similarity measure.

9.3 Rule-Based, CBR and Machine Learning: A Mixed Approach

Author: Asnel Christian (100%)

Although it remained purely theoretical, the following approach was also considered. In this approach, we attempt to programmatically generate the rules, thus eliminating the problem of the rule-based approach and use machine learning to the process thus allowing the process to be trainable.

This approach can be illustrated through the following steps:

- **Proximity/similarity:** In CBR, cases are stored in a case library and the system searches for similar cases to the current input based on their proximity or similarity. This is done to identify relevant cases that can inform the policy recommendation for the current input.
- **Decision tree:** Once the system has identified the nearest neighbors to the current input, a decision tree is built for each policy. The decision tree is a tool used to represent a set of rules and conditions that can be used to make a decision about which policy to recommend. By building a decision tree for each policy based on the nearest neighbor cases, the system is able to create a dynamic rule set that is specific to the problem and the input.

- **Class/policy determination:** Using the generated decision trees, the system is able to determine the class or policy to apply for the input. The decision tree allows the system to evaluate the conditions and *rules for each policy and determine which policy is most appropriate for the input.*

In this chapter, we discussed and showcased all the different approaches we considered to build out this recommendation system. Starting from a purely rule-based approach, to a more machine learning based approach. We highlighted each one of these approaches, so they may be considered by future research labs. In the next chapter, we have a discussion around this research lab, highlighting the scope of the research and provide recommendations and potential future work.

10. Discussion, Recommendation, and Research Scope

Author: Shehab (100%)

Analyzing the results from the previous chapters can provide an understanding of what has been achieved during this research lab as well as highlighting the missed opportunities, areas or improvements, scope, limitations, and future directions as well as recommendations for upcoming research.

Our research started with an idea to develop an expert system to help in combating Covid-19. While going through the literature and discussing results to find the right direction to drive this project into a specific distance, we experimented with different directions and ideas. We spent a considerable amount of time narrowing down to main questions that the expert system might answer. Among the several questions, one was around resource allocation and the other was around providing policy recommendations (strategy). We spent time on the topic of resource allocation however, it was deemed not suitable due to the lack of data that we required to build the expert system. Therefore, we continued with the second question because the resources and data required to build the system were available.

With this in mind, we experimented and discussed different approaches to build the system. Among these approaches that we tried are the following which are also mentioned in an earlier chapter:

- Using rule-based system only.
- Using several machine learning approaches to build the system.

However, we decided finally to focus on case-based reasoning due to the availability of data, and to explore new techniques of recommendation that can be generalized to similar type of pandemics because it relies on historical records and solutions. The system and the architecture we built recommends three types of policies, lockdown, mask and vaccine. We prepared data for this type of system, spent a good amount of time finding it and creating timelines for these cases. The result of this work is a system that is capable of recommending policies based on historical solutions that were adopted in Germany during Covid-19 from April 2020 to late 2022. In addition to this, we have adopted a rule-based approach to fine tune the system since the data collected captured periods that heavily influenced solutions and led to recommendations that were not good. For

example, in mid-2022, there were times where the number of active cases were high, specifically when we look at Rheinland-Pfalz, but low deaths. During this period there were no lockdown policies implemented as the vaccine had been already released. From this we can realize that the system is going to recommend no lockdown if the number of cases is high, because the cases stored did not have a situation where the number of cases were high and there was a lockdown policy in place. To combat issues like this, we had adopted a rule-based approach in a phase we refer to as case adaptation. Case adaptation is our way of telling the system not to generalize its answers to every scenario just because that is what it knows.

However, looking at what has been achieved and the different directions we explored initially, and as with anything, there is always a limitation. When it comes to the system in general, there are some areas that could be improved. Below are some points:

- Standardizing the data in case library on one scale for better similarity measurement.
- Experimenting with different similarity metrics such as Cosine Similarity because the system currently uses Euclidean distance.
- Introducing clustering as a different approach to find similar cases using a density-based clustering algorithm or other clustering techniques.
- Adding roles to the system to enforce authorization based on the user role.
- Improve the rule adaptation based on informed expert opinion.
- Add more cases from other cities in other states.
- Explore policy effectiveness as standalone research.

Since this research left some areas out of scope such as resource allocation and other implementations of this system, for future research, some of these areas can be explored more and compare the results with our approach or improve upon the current limitations. To summarize our research scope and recommendation, we focused mainly on CBR specifically on policy recommendation while we had already explored different options, but these could be tackled in a different research lab. We think using different approaches as recommended earlier can improve the current state of the system and introduce new research opportunities.

11. Conclusion

In conclusion, this research has successfully explored various themes and developed an architecture and a working prototype powered by case-based reasoning that compares past COVID-19 solutions to a new problem and provides policy recommendations for decision makers in similar pandemic context. The system addresses the research questions by offering optimal policies for future outbreaks based on past solutions. While limitations exist, such as data standardization and the need for role-based authorization as well as other factors, future research can explore enhancements and compare results as well as explore new themes or techniques. Overall, the developed system bridges the gap between AI and public health, aiding decision-making and contributing to the field of AI in healthcare.

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